When Spectra Forget Order:
Spectral Blindness, Observer-Marked Separation, and Stable Reconstruction of Ordered Matrix Products

Monty Dabas

July 25, 2026

Abstract

Ordered products retain factor order and intermediate evolution, while characteristic determinants retain only compressed endpoint information. We prove exact spectral-blindness results and construct finite observer measurements that restore endpoint distinctions with quantitative stability. For distinct labels, the square-free connected coefficient has the cyclic-descent form

$$\Phi_c^{(n)}(q) = -\frac{1}{(1-q)^n} \sum_e q^{\mathrm{cdes}_c(e)} T_e.$$

This yields complete four- and five-factor cyclic classifications and generic reconstruction transforms. For an arbitrary multiplicity partition $\lambda = (m_1, \ldots, m_r) \vdash n$, occurrence polarization gives the collapsed transform

$$\mathbf{\Phi}_\lambda(q) = -\frac{1}{(\prod_i m_i!)(1-q)^n} B_\lambda M_n(q) P_\lambda \mathbf{T}_\lambda,$$

where P_λ collapses distinct occurrences to multiset necklaces and B_λ restricts the source module to block words. This gives an exact all-spectral-value rank formula for every multiplicity partition. At degree five the collapsed hierarchy contains fifty formal necklaces but has rank forty-five, with a five-dimensional blind kernel. The defect is a source-restriction effect: the full split-occurrence transform $M_n(q)P_\lambda$ is generically complete because $M_n(q) = qI + O(q^2)$. Five explicit split source coefficients at $q = 2$ restrict to an invertible matrix of determinant -2592 on the blind kernel, proving that exactly five supplemental scalar observations are necessary and sufficient to repair the degree-five collapsed map. Observer-marker banks then provide endpoint separation and frame-stable reconstruction, while endpoint measurements still cannot reconstruct histories already collapsed to the same endpoint. All results are finite-dimensional; no automatic trace-class, Fredholm, hardware, or history-reconstruction extension is claimed.

Keywords: ordered matrix products; spectral blindness; cyclic descents; multiplicity partitions; split occurrences; multiset necklaces; connected log-determinant; observability defect; minimal observer bank; frame bounds; informational completeness; noisy reconstruction; Magnus expansion; target faithfulness.

Contents

1 Introduction

An ordered product of matrices carries several logically distinct layers of information. It has an endpoint matrix, a characteristic polynomial, a spectrum with algebraic multiplicities, and a factor history. When the factors do not commute, the factor history also determines commutator and nested-commutator terms in Baker–Campbell–Hausdorff or Magnus coordinates. These layers are frequently treated as interchangeable because the endpoint matrix determines its own spectrum and because spectral invariants are convenient to compute. The reverse implication fails: spectral data need not determine the endpoint, and the endpoint need not determine the factor history.

This paper develops a finite-dimensional theory for that information loss. The motivating question is simple:

> Which order distinctions survive spectral compression, and which additional declared observations are sufficient to recover the distinctions that ordinary spectral data erase?

The first answer is a no-go theorem. For invertible matrices X, Y, the two ordered products XY and YX are similar. Consequently,

$$\det(I - qXY) = \det(I - qYX)$$

for every scalar q, even when $XY \neq YX$. This exact identity is stronger than a perturbative cancellation. It says that no coefficient of the ordinary characteristic determinant can distinguish the two orders. The same mechanism makes every cyclic permutation of an invertible factor word spectrally indistinguishable.

The second answer is selective rather than entirely negative. In a three-factor problem, noncyclic permutations need not be similar. Passing to a connected log-determinant expansion reveals an order-sensitive cubic channel controlled by the cyclic trace form

$$\chi(A, B, C) = \mathrm{Tr}(A[B, C]).$$

Thus spectral compression does not erase every trace of order. It retains only conjugacy-invariant combinations that close inside trace moments.

The third answer introduces an observer. A marker M produces the response

$$\tau_M(U) = \mathrm{Tr}(MU).$$

A declared bank of markers is an analysis map on the matrix space. The bank may be complete, incomplete, well conditioned, or nearly blind. This elementary observation leads to a precise reconstruction theory. Matrix-unit markers recover every endpoint entry. Hermitian markers also form a complete bank. More generally, the frame operator of the bank controls injectivity, reconstruction error, pairwise discrimination under noise, and implementation perturbations.

The present paper is a mathematical successor to the author's earlier thermodynamic response work [1]. The earlier paper used curvature and cyclic return residue to organize memory in thermodynamic response. Here the physical interpretation is deliberately stripped away. We isolate the finite matrix problem, state the exact blindness and reconstruction theorems, and keep the scope within ordinary finite-dimensional linear algebra, trace identities, and frame theory. This separation is useful both scientifically and expository: the mathematics can be evaluated without accepting a larger interpretive framework.

1.1 Main contributions

The results are organized around six claims.

C1. Exact spectral blindness. Two-factor order and cyclic position are invisible to every characteristic determinant of the endpoint product.

C2. First generic noncyclic channel. For three labelled exponential factors, the first multilinear connected determinant difference is proportional to $\mathrm{Tr}(A[B,C])$.

C3. Observer-marked endpoint separation. A marker detects an endpoint difference precisely through a nonzero trace pairing. Complete finite banks reconstruct every endpoint difference.

C4. Quantitative stability. Frame bounds give sharp noisy reconstruction and discrimination estimates, along with perturbation stability.

C5. Optimal bank geometry. At fixed total marker energy, tight banks maximize the worst-direction response and minimize the analysis condition number.

C6. Blind-kernel and history boundaries. Endpoint reconstruction, target preservation, and factor-history reconstruction are different tasks. No endpoint-only bank can recover histories that produce the same endpoint.

1.2 Why the negative results matter

A negative theorem is not merely a warning against a poor feature choice. It determines what no downstream algorithm can repair from the same compressed data. If two endpoints have the same ordinary characteristic determinant, no post-processing of that determinant can recover their order. If two factor histories produce the same endpoint, no endpoint-only marker bank, however complete on matrix space, can recover the erased history. These are information-theoretic boundaries.

Conversely, declaring an observer bank changes the representation itself. The recovery theorem is therefore not the claim that a clever formula extracts hidden order from an unchanged spectrum. It is the claim that additional linear measurements make a previously blind representation faithful on a declared class of endpoint residues.

1.3 Relation to standard literature

The ordered-exponential viewpoint is naturally connected to the Magnus expansion [2, 3]; the matrix analysis and similarity arguments are standard [4, 5]; and the quantitative observer theory is a finite frame problem in the tradition initiated by Duffin and Schaeffer [11, 12]. The novelty of the present synthesis lies in treating these ingredients as one explicit blindness-and-recovery chain, including the endpoint/history distinction and the sharp finite marker-conditioning results.

1.4 Scope

All core theorems are finite-dimensional. We do not silently replace matrices by unbounded operators, ordinary determinants by Fredholm determinants, or finite frame bounds by infinite-dimensional closed-range statements. Extensions to trace ideals, analytic Fredholm pencils, continuous-time observation, symmetry-constrained design, and nonlinear history reconstruction are left as open problems.

2 Notation and problem formulation

Let $V \cong \mathbb{C}^d$, and write

$$\mathrm{M}_d(\mathbb{C}) = \mathrm{End}(V).$$

For $D \in \mathrm{M}_d(\mathbb{C})$, the Hilbert–Schmidt inner product and norm are

$$\langle M, D\rangle_{\mathrm{HS}} = \mathrm{Tr}(M^* D), \qquad \|D\|_{\mathrm{HS}}^2 = \mathrm{Tr}(D^* D).$$

The ordinary trace pairing $\mathrm{Tr}(MD)$ is recovered by inserting M^* in the Hilbert–Schmidt convention.

2.1 Ordered products

Let $A_1, \dots, A_n \in \mathrm{M}_d(\mathbb{C})$ be labelled generators and let

$$X_j(z_j) = e^{z_j A_j}.$$

For a permutation π of $\{1, \dots, n\}$, define

$$U_\pi(\mathbf{z}) = X_{\pi(n)}(z_{\pi(n)}) \cdots X_{\pi(1)}(z_{\pi(1)}).$$

We distinguish three objects:

(i) the *word* or ordered factor history π;

(ii) the *endpoint* U_π;

(iii) a *spectral shadow*, such as $\det(I - qU_\pi)$.

The maps

$$\pi \longmapsto U_\pi \longmapsto \det(I - qU_\pi)$$

can each be many-to-one.

Definition 2.1 (Endpoint collision)**.** *Two factor histories π, σ have an endpoint collision if*

$$U_\pi = U_\sigma$$

although the factor words differ.

Definition 2.2 (Spectral collision)**.** *Two endpoints U, V have a characteristic-determinant collision if*

$$\det(I - qU) = \det(I - qV)$$

as polynomials in q.

Endpoint collision implies spectral collision, but the converse fails.

2.2 Normalized connected determinant

For an endpoint $U_\pi(\mathbf{z})$, define

$$F_\pi(q,\mathbf{z}) = \frac{\det(I - qU_\pi(\mathbf{z}))}{(1-q)^d}, \qquad L_\pi(q,\mathbf{z}) = \log F_\pi(q,\mathbf{z}).$$

The normalization gives $F_\pi(q,\mathbf{0}) = 1$. We may treat q formally. Analytically, the familiar trace-log expansion is valid in any regime where the logarithm is defined and the corresponding series converges.

2.3 Observer banks

A marker bank is an ordered finite family

$$\mathcal{M} = (M_a)_{a=1}^r \subset \mathrm{M}_d(\mathbb{C}).$$

Its analysis map is

$$\mathcal{A}_\mathcal{M} : \mathrm{M}_d(\mathbb{C}) \to \mathbb{C}^r, \qquad \mathcal{A}_\mathcal{M}(D) = \big(\langle M_a, D\rangle_{\mathrm{HS}}\big)_{a=1}^r.$$

For endpoints U, V, the observer residue is

$$G_\mathcal{M}(U,V) = \mathcal{A}_\mathcal{M}(U - V).$$

Definition 2.3 (Complete bank)**.** *The bank $\mathcal{M}$ is complete on a subspace $\mathcal{D} \subseteq \mathrm{M}_d(\mathbb{C})$ if*

$$\ker \mathcal{A}_\mathcal{M} \cap \mathcal{D} = \{0\}.$$

It is complete on the whole matrix space when $\ker \mathcal{A}_\mathcal{M} = \{0\}$.

Completeness is relative to the declared residue class. A small bank may be complete on a low-dimensional model class while being blind on the ambient matrix space.

2.4 Three recovery objectives

It is useful to distinguish three target questions.

R1. Endpoint separation: determine whether $U = V$.

R2. Target preservation: determine a selected target $P(U)$ without reconstructing all of U.

R3. History reconstruction: recover the original factor word or path.

The observer-bank theorems solve the first problem in finite dimension and give exact criteria for the second. They do not solve the third in general.

3 Determinant compression and connected spectral channels

The characteristic determinant is a conjugacy invariant. It is therefore a compression of the endpoint matrix, and the endpoint matrix is itself a compression of the factor history.

Proposition 3.1 (Trace-moment representation)**.** *Let $U \in \mathrm{M}_d(\mathbb{C})$. Formally in q,*

$$\log\det(I - qU) = -\sum_{m\ge 1} \frac{q^m}{m} \operatorname{Tr}(U^m).$$

Consequently, the characteristic determinant retains exactly the finite-dimensional spectral trace moments of U.

Proof. Use
$$\log\det(I-qU)=\operatorname{Tr}\log(I-qU)$$
and expand the matrix logarithm formally:
$$\log(I-qU)=-\sum_{m\geq 1}\frac{q^m}{m}U^m.$$
Taking the trace gives the result. □

If $U=e^{\Omega}$ for a chosen logarithm Ω, then the normalized connected determinant satisfies
$$L(q)=-\sum_{m\geq 1}\frac{q^m}{m}\big(\operatorname{Tr}(e^{m\Omega})-d\big).$$

The Magnus or Baker–Campbell–Hausdorff generator Ω may contain many commutators and nested commutators. The determinant retains only combinations that survive inside the traces $\operatorname{Tr}(e^{m\Omega})$.

Definition 3.2 (Spectral loss map)**.** *The map*
$$\mathfrak{S}:\mathrm{M}_d(\mathbb{C})\to\mathbb{C}[q],\qquad \mathfrak{S}(U)=\det(I-qU)$$
is the characteristic spectral loss map.

The fibre $\mathfrak{S}^{-1}(\mathfrak{S}(U))$ contains every matrix with the same characteristic polynomial as U, including all matrices similar to U, and may contain nonsimilar matrices as well.

3.1 Connected versus disconnected coefficients

Suppose
$$V(\mathbf{z})=\sum_a z_aV_a,\qquad F(E,\mathbf{z})=\det(I-G(E)V(\mathbf{z})).$$
Writing
$$L(E,\mathbf{z})=\log F(E,\mathbf{z})$$
removes products of lower-order contributions from the primitive coefficient ledger. For example, if
$$\kappa_i=[z_i]L,\qquad \kappa_{ij}=[z_iz_j]L,$$
then the ordinary determinant coefficient has the moment–cumulant form
$$[z_iz_j]F=\kappa_{ij}+\kappa_i\kappa_j.$$
Thus a high-order determinant monomial is not automatically a genuinely new interaction. The log-determinant isolates connected trace channels.

For the present paper, connected coefficients are used only to identify the first noncyclic order-sensitive spectral term. No infinite seam tower or renormalized determinant is assumed.

4 Position relative to classical literature and novelty boundary

The purpose of this section is to separate the classical ingredients used in the paper from the claims made by the present synthesis. This distinction is mathematically important. A standard identity does not become new because it is renamed, while a new theorem may consist of a previously unrecorded comparison, filtration, sharp constant, or combination of established tools.

4.1 Determinantal identities and cyclic blindness

The identity

$$\det(I_m + AB) = \det(I_n + BA), \qquad A \in \mathbb{C}^{m\times n}, \quad B \in \mathbb{C}^{n\times m},$$

is the classical Sylvester determinant theorem; see, for example, [4]. In the square case it implies

$$\det(I - qXY) = \det(I - qYX)$$

without any invertibility assumption. When one factor is invertible, the two products are additionally similar. The paper therefore does not claim the bare two-factor determinant identity as historically new. Its role here is to serve as the first exact blindness gate in an information hierarchy:

$$\text{ordered word} \longrightarrow \text{endpoint matrix} \longrightarrow \text{characteristic spectral shadow}.$$

Iterating Sylvester's identity also gives characteristic-determinant invariance under cyclic rotation of a square factor word, again without requiring every factor to be invertible. Similarity of the cyclic rotations is the stronger special case available when the appropriate factor is invertible.

4.2 Trace words and simultaneous conjugation

Trace words are central objects in matrix invariant theory. In characteristic zero, polynomial invariants of several matrices under simultaneous conjugation are generated by traces of matrix words; Procesi's foundational treatment is [6]. This classical background explains why a trace-closed quantity such as

$$\mathrm{Tr}(A[B, C])$$

is a natural conjugacy-invariant order channel. The present paper does not claim that traces of words, cyclic trace reduction, or simultaneous-conjugation invariance are new.

The specific result used here is more narrowly formulated. For the declared comparison

$$U_{CBA} = e^{zC}e^{yB}e^{xA}, \qquad U_{CAB} = e^{zC}e^{xA}e^{yB},$$

we identify the lowest total-degree multilinear coefficient in the difference of normalized connected log determinants and prove that it is

$$[xyz](L_{CBA} - L_{CAB}) = \frac{q}{(1-q)^2}\,\mathrm{Tr}(A[B, C]).$$

The phrase "first noncyclic channel" in this paper refers to this explicit degree filtration for the stated comparison. It is not a claim that cyclic trace forms were absent from the earlier invariant-theory literature.

4.3 Ordered exponentials and Magnus coordinates

The representation of a time-ordered linear evolution by an exponential whose logarithm contains commutators and nested commutators originates with Magnus [2]; a broad review is [3]. Baker–Campbell–Hausdorff and Magnus coordinates describe information carried by the logarithm of an ordered endpoint. The present contribution is not a new Magnus expansion. Instead, it asks which of those order-dependent terms survive the further compression

$$U \longmapsto \det(I - qU).$$

The answer is selective: pair-order information may be present in the logarithm while being annihilated by the characteristic determinant, whereas certain higher trace closures survive.

4.4 Frames, informational completeness, and tomography

The quantitative marker-bank results belong to finite frame theory, whose classical origins include Duffin and Schaeffer [11]; see also [12]. Closely related operator-reconstruction ideas appear in informationally complete quantum measurements, where frame methods are used to reconstruct operators from expectation data [13]. Tight informationally complete measurements and their reconstruction advantages are developed in [14].

Accordingly, this paper does not claim the general frame inequality, canonical dual reconstruction, pseudoinverse stability, or tight-frame optimality as new facts of frame theory. What is contributed is their specialization to a precisely declared order-blindness problem, together with a common notation and sharp deterministic thresholds that connect endpoint residues to observer responses.

4.5 Contribution matrix

Ingredient	Classical status	Role in this paper
Sylvester determinant identity	Classical matrix identity	Exact two-factor and cyclic characteristic-blindness gate, strengthened to singular factors.
Trace cyclicity and trace-word invariants	Classical invariant theory	Identifies the conjugacy-invariant codomain in which a noncyclic order channel can survive.
Magnus/BCH commutator expansion	Classical ordered-exponential theory	Supplies order-memory coordinates whose loss under determinant compression is classified.
Finite frame bounds and canonical duals	Classical frame theory	Quantifies completeness, noise amplification, pairwise detection, and marker perturbations for endpoint residues.
Informational completeness and tight measurement design	Established in operator tomography	Provides the closest neighbouring reconstruction paradigm and clarifies that our markers are general matrix probes, not automatically positive quantum effects.
Endpoint/history distinction	Elementary in principle, often left implicit	Promoted to an explicit no-go boundary: endpoint-complete observations cannot recover distinct histories with the same endpoint.

4.6 Conservative originality statement

The paper's originality claim is therefore deliberately composite and limited. It consists of:

(i) one explicit hierarchy separating word, endpoint, spectral shadow, marked endpoint response, and protected target;

(ii) a degree-resolved calculation of the cubic connected order coefficient for the declared three-factor comparison;

(iii) a single theorem chain from exact spectral blindness to complete marked endpoint recovery, sharp deterministic noise bounds, perturbation stability, and tight-bank design;

(iv) a blind-kernel criterion that separates endpoint reconstruction, target faithfulness, and impossible history reconstruction.

No claim is made that every constituent identity is historically new, nor that this literature audit is an exhaustive priority search. The theorem statements are intended to be independently checkable even if a closer precursor is later identified for one constituent calculation.

5 Exact spectral blindness

5.1 Two-factor order

Theorem 5.1 (Exact two-factor characteristic blindness). *Let $X, Y \in \mathrm{M}_d(\mathbb{C})$ be arbitrary. Then*

$$\boxed{\det(I - qYX) = \det(I - qXY)}$$

for every scalar q.

Proof. Sylvester's determinant identity states that

$$\det(I_m + AB) = \det(I_n + BA)$$

for $A \in \mathbb{C}^{m\times n}$ and $B \in \mathbb{C}^{n\times m}$. Apply it in the square case with $A = -qY$ and $B = X$. □

Proposition 5.2 (Similarity special case). *If X is invertible, then*

$$YX = X^{-1}(XY)X.$$

Thus invertibility strengthens characteristic-polynomial equality to similarity, but it is not required for definition 5.1.

The endpoints can nevertheless differ:

$$YX - XY = [Y, X] \neq 0.$$

Hence the characteristic determinant can be exactly blind to a nonzero endpoint residue.

Corollary 5.3 (Exponential two-factor blindness). *For arbitrary $A, B \in \mathrm{M}_d(\mathbb{C})$ and $x, y \in \mathbb{C}$,*

$$\det(I - qe^{yB}e^{xA}) = \det(I - qe^{xA}e^{yB}).$$

No convergence of a Magnus series is involved. The equality follows directly from definition 5.1; the exponential factors are additionally invertible, so the two products are similar.

5.2 Cyclic permutation blindness

Theorem 5.4 (Cyclic-word characteristic blindness). *Let $X_1, \ldots, X_n \in \mathrm{M}_d(\mathbb{C})$ be arbitrary square matrices. Then every cyclic rotation of the product*

$$U = X_n X_{n-1} \cdots X_1$$

has the same characteristic determinant. In particular,

$$\det(I - qX_n \cdots X_1) = \det(I - qX_{n-1} \cdots X_1 X_n).$$

If all factors are invertible, the cyclic rotations are also similar.

Proof. Write $U = X_n Q$, where $Q = X_{n-1} \cdots X_1$. Sylvester's identity gives

$$\det(I - qX_n Q) = \det(I - qQX_n).$$

Repeating the argument proves equality for every cyclic rotation. If X_n is invertible, then $QX_n = X_n^{-1} U X_n$; iterating gives similarity when all factors are invertible. □

Remark 5.5. *The characteristic determinant treats a square factor word as a cyclic word, or necklace, rather than a linear word. A cyclic origin is not observable from the characteristic determinant alone. For singular factors the conclusion is characteristic-polynomial equality, not necessarily similarity.*

5.3 Blindness is stronger than trace blindness

For every pair X, Y,

$$\mathrm{Tr}(YX - XY) = 0.$$

Trace blindness is therefore automatic for a commutator. Definition 5.1 is substantially stronger: it identifies equality of the entire characteristic polynomial, not merely the first trace moment.

5.4 Nonsufficiency of noncommutation

The condition $[X, Y] \neq 0$ proves that the endpoints differ, but it does not select a visible spectral channel. Two-factor characteristic data remain blind regardless of the size of the commutator. Likewise, in higher-factor problems, noncommutation creates possible order-sensitive trace forms but does not force any prescribed form to be nonzero.

6 The lowest-degree noncyclic spectral order channel

Two-factor reversal and cyclic permutation are spectrally invisible. For the declared three-factor comparison below, the lowest total-degree multilinear connected channel is obtained when a third labelled generator closes a pair commutator inside the trace.

Let

$$U_{CBA}(x, y, z) = e^{zC}e^{yB}e^{xA}, \qquad U_{CAB}(x, y, z) = e^{zC}e^{xA}e^{yB},$$

and define

$$L_{CBA} = \log \frac{\det(I - qU_{CBA})}{(1-q)^d}, \qquad L_{CAB} = \log \frac{\det(I - qU_{CAB})}{(1-q)^d}.$$

Theorem 6.1 (Cubic noncyclic order coefficient). *The multilinear coefficient satisfies*

$$\boxed{[xyz]\big(L_{CBA} - L_{CAB}\big) = \frac{q}{(1-q)^2}\,\mathrm{Tr}(A[B, C]).}$$

A proof by direct trace expansion is given in section A. The result does not require convergence of the full Magnus series because only a finite formal multilinear coefficient is extracted.

Definition 6.2 (Cyclic three-form). *Define*

$$\chi(A, B, C) = \mathrm{Tr}(A[B, C]).$$

By cyclicity of trace,

$$\chi(A, B, C) = \mathrm{Tr}(B[C, A]) = \mathrm{Tr}(C[A, B]).$$

Interchanging two arguments changes its sign.

Corollary 6.3 (Visibility criterion). *The two noncyclic orderings CBA and CAB are distinguished by the cubic connected spectral channel whenever*

$$\chi(A, B, C) \neq 0.$$

Remark 6.4 (Noncommutation is not sufficient). *It is possible to have $[B, C] \neq 0$ while*

$$\mathrm{Tr}(A[B, C]) = 0.$$

Thus the commutator is an available order-memory direction, whereas spectral visibility requires a nonzero closure against the third labelled generator.

6.1 Interpretation through Magnus coordinates

The logarithm of an ordered product contains pair commutators at quadratic order and nested commutators at higher order. For two factors, reversing the order changes the logarithm but the characteristic determinant annihilates that distinction. For three factors, a third generator can pair with the commutator through a cyclic trace:

$$\text{pair-order memory} + \text{third labelled step} \longrightarrow \text{connected cyclic spectral response.}$$

The theorem identifies a quotient of order memory, not a reconstruction of the full factor word.

7 Observer-marked separation

Ordinary characteristic data are invariant under similarity. To recover endpoint distinctions erased by that quotient, we enlarge the observation map.

Definition 7.1 (Marked trace)**.** *For a marker $M \in \mathrm{M}_d(\mathbb{C})$, define*

$$\tau_M(U) = \mathrm{Tr}(MU).$$

For endpoints U, V, the marked residue is

$$G_M(U, V) = \mathrm{Tr}(M(U - V)).$$

Theorem 7.2 (Marked detection of two-factor reversal)**.** *Let*

$$U_{YX} = YX, \qquad U_{XY} = XY.$$

Then

$$G_M(U_{YX}, U_{XY}) = \mathrm{Tr}(M[Y, X]).$$

Equivalently,

$$G_M(U_{YX}, U_{XY}) = \mathrm{Tr}([M, Y]X).$$

Therefore

$$\mathrm{Tr}(M[Y, X]) \neq 0$$

is sufficient for M to detect the order reversal.

Proof. Substitute $U_{YX} - U_{XY} = YX - XY = [Y, X]$. The alternative form follows by cyclicity:

$$\mathrm{Tr}(MYX - MXY) = \mathrm{Tr}(MYX - YMX) = \mathrm{Tr}([M, Y]X).$$

□

A prescribed marker may still be blind even when $[Y, X] \neq 0$. The condition is not noncommutation alone but nonzero pairing with the declared marker.

7.1 Marked determinants

Define

$$F_M(q; U) = \det(I - qMU).$$

Lemma 7.3 (Linear marked coefficient)**.**

$$[q]F_M(q; U) = -\,\mathrm{Tr}(MU).$$

Proof. Use the first variation

$$\det(I + tK) = 1 + t\,\mathrm{Tr}(K) + O(t^2)$$

with $t = -q$ and $K = MU$. □

Theorem 7.4 (Marked determinant separation)**.** *If*

$$\mathrm{Tr}(M(U - V)) \neq 0,$$

then

$$F_M(q; U) \neq F_M(q; V)$$

as polynomials in q.

The marked determinant does not extract order from the ordinary determinant. It is a different observation because the marker is inserted before spectral compression.

7.2 Similarity covariance and observer relativity

If S is invertible and both endpoint and marker are transported,

$$U \mapsto SUS^{-1}, \qquad M \mapsto SMS^{-1},$$

then

$$\mathrm{Tr}(SMS^{-1}SUS^{-1}) = \mathrm{Tr}(MU)$$

and

$$\det(I - qSMS^{-1}SUS^{-1}) = \det(I - qMU).$$

Proposition 7.5 (Simultaneous covariance)**.** *Marked traces and marked determinants are invariant under simultaneous similarity transport of the endpoint and observer frame.*

If the endpoint is conjugated while M is held fixed, the response may change. A fixed marker is therefore observer-relative. Covariance of a transported observer frame is not the same as invariance of one fixed marker.

8 Four-factor adjacent-swap channels

The cubic result in definition 6.1 contains one spectator generator closing a pair commutator inside a cyclic trace. With four labelled factors, an adjacent swap has two spectator generators. Their order matters, and the multilinear connected response splits into two context channels.

Let

$$U_{DCBA}(w, z, y, x) = e^{wD}e^{zC}e^{yB}e^{xA}, \qquad U_{DCAB}(w, z, y, x) = e^{wD}e^{zC}e^{xA}e^{yB}.$$

Define the normalized connected determinants

$$L_{DCBA} = \log \frac{\det(I - qU_{DCBA})}{(1 - q)^d}, \qquad L_{DCAB} = \log \frac{\det(I - qU_{DCAB})}{(1 - q)^d}.$$

The variables x, y, z, w label the four generators, while q remains the spectral variable.

Theorem 8.1 (Four-factor adjacent-swap coefficient)**.** *The complete multilinear coefficient of the adjacent transposition $BA \mapsto AB$ is*

$$\boxed{[xyzw]\big(L_{DCBA} - L_{DCAB}\big) = \frac{q}{(1-q)^3}\operatorname{Tr}\!\big(A[B,DC]\big) + \frac{q^2}{(1-q)^3}\operatorname{Tr}\!\big(A[B,CD]\big).}$$

Equivalently,

$$\boxed{[xyzw]\big(L_{DCBA} - L_{DCAB}\big) = \frac{q(1+q)}{2(1-q)^3}\operatorname{Tr}\!\big(A[B,\{C,D\}]\big) - \frac{q}{2(1-q)^2}\operatorname{Tr}\!\big(A[B,[C,D]]\big).}$$

Here $\{C,D\} = CD + DC$.

Proof. Write

$$r = \frac{q}{1-q}, \qquad U_j = I + W_j.$$

Then

$$L_j = \operatorname{Tr}\log(I - rW_j) = -\sum_{k\geq 1}\frac{r^k}{k}\operatorname{Tr}(W_j^k).$$

Only $k = 1,2,3,4$ can contribute to the coefficient $[xyzw]$. Expanding both ordered products through multilinear degree four and reducing trace words cyclically gives

$$\begin{aligned}[xyzw](L_{DCBA} - L_{DCAB}) &= r(1+r)^2\big(\operatorname{Tr}(ABDC) - \operatorname{Tr}(ADCB)\big)\\ &\quad + r^2(1+r)\big(\operatorname{Tr}(ABCD) - \operatorname{Tr}(ACDB)\big).\end{aligned}$$

The two trace differences are

$$\operatorname{Tr}(ABDC) - \operatorname{Tr}(ADCB) = \operatorname{Tr}\!\big(A[B,DC]\big),$$

$$\operatorname{Tr}(ABCD) - \operatorname{Tr}(ACDB) = \operatorname{Tr}\!\big(A[B,CD]\big).$$

Since

$$r(1+r)^2 = \frac{q}{(1-q)^3}, \qquad r^2(1+r) = \frac{q^2}{(1-q)^3},$$

the first formula follows.

For the second formula, use

$$DC = \frac{1}{2}\big(\{C,D\} - [C,D]\big), \qquad CD = \frac{1}{2}\big(\{C,D\} + [C,D]\big).$$

Collecting the symmetric and antisymmetric context terms yields the stated Jordan–Lie decomposition. $\square$

8.1 Two context channels

The theorem identifies two distinct ways in which the spectator pair (C,D) can make the swap of (A,B) visible.

Definition 8.2 (Symmetric-context channel)**.**

$$\mathcal{J}_4(A,B;C,D) = \operatorname{Tr}\!\big(A[B,\{C,D\}]\big).$$

Definition 8.3 (Nested-commutator channel)**.**

$$\mathcal{L}_4(A,B;C,D) = \operatorname{Tr}\!\big(A[B,[C,D]]\big).$$

Thus

$$[xyzw](L_{DCBA} - L_{DCAB}) = K_J(q)\,\mathcal{J}_4(A,B;C,D) + K_L(q)\,\mathcal{L}_4(A,B;C,D),$$

where

$$K_J(q) = \frac{q(1+q)}{2(1-q)^3}, \qquad K_L(q) = -\frac{q}{2(1-q)^2}.$$

The two rational kernels are not scalar multiples. Four-factor adjacent-swap visibility therefore has two independent spectral response profiles.

Corollary 8.4 (Commuting spectators)**.** *If $[C,D]=0$, then the nested-commutator channel vanishes and*

$$[xyzw](L_{DCBA} - L_{DCAB}) = \frac{q(1+q)}{2(1-q)^3}\mathcal{J}_4(A,B;C,D).$$

Corollary 8.5 (Anticommuting spectators)**.** *If $\{C,D\}=0$, then the symmetric-context channel vanishes and*

$$[xyzw](L_{DCBA} - L_{DCAB}) = -\frac{q}{2(1-q)^2}\mathcal{L}_4(A,B;C,D).$$

Corollary 8.6 (Degree-four blindness criterion)**.** *The declared adjacent swap is invisible in the complete multilinear four-factor connected channel exactly when*

$$q\ \mathrm{Tr}\big(A[B,DC]\big) + q^2\ \mathrm{Tr}\big(A[B,CD]\big) = 0$$

as a rational function of q. Equivalently, both context traces vanish:

$$\mathrm{Tr}\big(A[B,DC]\big) = 0, \qquad \mathrm{Tr}\big(A[B,CD]\big) = 0.$$

This criterion concerns degree four only; higher repeated-variable channels may remain nonzero.

8.2 Cyclic normal form for adjacent transpositions

There are six cyclic word classes containing four distinct labels. The trace forgets the choice of cyclic origin, so any adjacent transposition in a four-factor word can be rotated until the exchanged pair occupies the two rightmost positions. Relabelling the two spectators then gives the normal form of definition 8.1.

Proposition 8.7 (Adjacent-swap cyclic normal form)**.** *Every multilinear connected difference produced by one adjacent transposition of four labelled exponential factors is obtained from definition 8.1 by a cyclic relabelling of A,B,C,D. Consequently, every such difference decomposes into one symmetric-context trace and one nested-commutator trace.*

Proof. Characteristic determinants, and hence their connected logarithms, are invariant under cyclic rotation of the factor word. Rotate the word until the exchanged adjacent pair is rightmost, call it (B,A), and call the remaining ordered spectators (D,C). Apply definition 8.1. □

This is a classification of adjacent transpositions, not of every difference between arbitrary four-factor permutations. A general permutation may be decomposed into adjacent transpositions, but the connected responses of the intermediate words need not form an order-independent scalar ledger without a separate cocycle theorem.

8.3 Independent channel calibrations

The two context channels are algebraically independent in the sense that each can be nonzero while the other vanishes.

Example 8.8 (Symmetric context without nested context)**.** *In* $\mathrm{M}_2(\mathbb{C})$*, take*

$$A = E_{12}, \qquad B = E_{21}, \qquad C = D = E_{11}.$$

Then $[C, D] = 0$*, while*

$$\mathcal{J}_4(A, B; C, D) = 2, \qquad \mathcal{L}_4(A, B; C, D) = 0.$$

The complete coefficient is

$$\frac{q(1+q)}{(1-q)^3}.$$

Example 8.9 (Nested context without symmetric context)**.** *In* $\mathrm{M}_2(\mathbb{C})$*, take*

$$A = C = E_{12}, \qquad B = D = E_{21}.$$

Then $\{C, D\} = I$*, which commutes with* B*, so*

$$\mathcal{J}_4(A, B; C, D) = 0, \qquad \mathcal{L}_4(A, B; C, D) = 2.$$

The complete coefficient is

$$-\frac{q}{(1-q)^2}.$$

8.4 Exact four-cycle calibration

Let

$$A = E_{12}, \qquad B = E_{23}, \qquad C = E_{34}, \qquad D = E_{41}$$

in $\mathrm{M}_4(\mathbb{C})$. Since each generator squares to zero, the exponentials truncate in the multilinear calculation. Direct determinant evaluation gives

$$\det(I - qU_{DCBA}) = (1-q)^4 - q^3 xyzw,$$
$$\det(I - qU_{DCAB}) = (1-q)^4 - q^2 xyzw.$$

Therefore

$$[xyzw](L_{DCBA} - L_{DCAB}) = \frac{q^2}{(1-q)^3}.$$

For the trace channels,

$$\mathrm{Tr}\big(A[B, DC]\big) = 0, \qquad \mathrm{Tr}\big(A[B, CD]\big) = 1,$$

and hence

$$\mathcal{J}_4(A, B; C, D) = 1, \qquad \mathcal{L}_4(A, B; C, D) = 1.$$

Both forms of definition 8.1 reproduce the exact coefficient.

8.5 Relation to cyclic words and invariant theory

The six multilinear trace classes at word length four belong to the classical trace-word invariant algebra [6]. Circular-word methods provide a natural combinatorial language for quotienting ordinary words by cyclic rotation [8]. The contribution here is not the existence of trace words or necklaces. It is the explicit connected-log-determinant weight assigned to an adjacent transposition and its decomposition into symmetric and nested-commutator context channels.

The next combinatorial gate is to classify arbitrary four-factor permutation differences in the six-dimensional cyclic-word space and determine which relations are generated by adjacent-swap channels with path-independent coefficients.

9 Six-class cyclic-word potential and path independence

The adjacent-swap calculation of section 8 can be completed at the level of all four-factor order classes. With four distinct labels, ordinary trace identifies words under cyclic rotation, leaving six oriented cyclic classes. Reversal pairs these six classes into three antipodal pairs.

Let

$$\mathcal{C}_4 = \{[ABCD], [ABDC], [ACBD], [ACDB], [ADBC], [ADCB]\}_{\text{cyc}},$$

and let

$$\rho : \mathcal{C}_4 \to \mathcal{C}_4$$

be reversal. Explicitly,

$$[ABCD] \leftrightarrow [ADCB], \qquad [ABDC] \leftrightarrow [ACDB], \qquad [ACBD] \leftrightarrow [ADBC].$$

For $c \in \mathcal{C}_4$, write

$$T_c = \mathrm{Tr}(W_c)$$

for the corresponding cyclic trace word, and let U_c denote any exponential factor word representing the class c, with one independent scalar variable attached to each of A, B, C, D. Cyclic determinant invariance makes the following coefficient independent of the chosen cyclic representative:

$$\Phi_c(q) = [xyzw] \log \frac{\det(I - qU_c)}{(1-q)^d}.$$

9.1 Universal six-class coefficient

Lemma 9.1 (Multilinear cut-count table)**.** *Fix $c \in \mathcal{C}_4$ and set $r = q/(1-q)$. In the expansion*

$$-\sum_{k=1}^{4} \frac{r^k}{k} \mathrm{Tr}(W^k),$$

the coefficient multiplying a cyclic trace class $e \in \mathcal{C}_4$ is

	r	r^2	r^3	r^4
$e = c$	-1	-3	-3	-1
$e = \rho(c)$	0	0	-1	-1
$e \notin \{c, \rho(c)\}$	0	-1	-2	$-1.$

Proof. The multilinear part of $U_c - I$ contains exactly one word for every nonempty subsequence of the chosen factor order. A contribution to $\mathrm{Tr}(W^k)$ is therefore an ordered decomposition of the four labels into k nonempty subsequences, followed by cyclic reduction of the concatenated word. Enumerating these ordered decompositions gives the three rows displayed above. The factor $1/k$ removes the k cyclic choices of the first block. No analytic convergence statement is used. $\square$

Theorem 9.2 (Six-class cyclic potential)**.** *For every $c \in \mathcal{C}_4$,*

$$\boxed{\Phi_c(q) = -\frac{1}{(1-q)^4}\left(qT_c + q^3 T_{\rho(c)} + q^2 \sum_{e \notin \{c,\rho(c)\}} T_e\right).}$$

Equivalently, with

$$T_{\text{all}} = \sum_{e \in \mathcal{C}_4} T_e,$$

one has

$$\boxed{\Phi_c(q) = -\frac{q^2}{(1-q)^4} T_{\text{all}} - \frac{q}{(1-q)^3} T_c + \frac{q^2}{(1-q)^3} T_{\rho(c)}.}$$

Proof. By definition 9.1, the own class has coefficient

$$-r(1+r)^3 = -\frac{q}{(1-q)^4},$$

the reversed class has coefficient

$$-r^3(1+r) = -\frac{q^3}{(1-q)^4},$$

and every remaining class has coefficient

$$-r^2(1+r)^2 = -\frac{q^2}{(1-q)^4}.$$

This gives the first formula. The second follows by adding and subtracting the common $q^2 T_{\text{all}}$ contribution. □

The theorem classifies the complete multilinear connected coefficient for all 24 linear permutations, because those permutations collapse to the six classes in $\mathcal{C}_4$.

9.2 Arbitrary permutation differences

Corollary 9.3 (Universal difference formula)**.** *For any $c, d \in \mathcal{C}_4$,*

$$\boxed{\Phi_c(q) - \Phi_d(q) = \frac{q}{(1-q)^3}\big(T_d - T_c\big) + \frac{q^2}{(1-q)^3}\big(T_{\rho(c)} - T_{\rho(d)}\big).}$$

Thus every arbitrary four-factor order difference is determined by two oriented trace-word differences: the direct pair and the reversed pair.

Corollary 9.4 (Reversal channel)**.** *For $d = \rho(c)$,*

$$\Phi_c(q) - \Phi_{\rho(c)}(q) = \frac{q(1+q)}{(1-q)^3}\big(T_{\rho(c)} - T_c\big).$$

The local Jordan–Lie formula of definition 8.1 is recovered by applying definition 9.3 to adjacent cyclic classes and rewriting the resulting trace-word differences as commutator closures.

9.3 Octahedral adjacent-swap graph

Let G_4 be the graph whose vertices are $\mathcal{C}_4$, with an edge whenever one class is obtained from the other by one adjacent transposition after a cyclic rotation.

Theorem 9.5 (Octahedral order graph)**.** *The graph G_4 is the octahedral graph. Every vertex has degree four, and the unique non-neighbour of c, apart from c itself, is its reversal $\rho(c)$. Equivalently,*

$$G_4 = K_6 \setminus \big\{\{c, \rho(c)\} : c \in \mathcal{C}_4\big\}.$$

Proof. A cyclic class has four adjacent pairs, including the pair crossing the chosen cyclic origin. Swapping any one of them produces four distinct cyclic classes. A single adjacent swap cannot reverse the orientation of the entire four-cycle, so $\rho(c)$ is not reached. Since only six vertices exist, the remaining four classes are exactly the neighbours of c. □

Consequently every non-reversal difference is one edge, while every reversal difference has graph distance two.

9.4 Exact edge cocycle and vanishing path residue

Orient an edge from c to d and define

$$\omega(c, d) = \Phi_c - \Phi_d.$$

Theorem 9.6 (Exact adjacent-swap cocycle)**.** *The edge response ω is the coboundary of the vertex potential Φ. Hence for every path*

$$c_0, c_1, \dots, c_m$$

in G_4,

$$\boxed{\sum_{j=1}^{m} \omega(c_{j-1}, c_j) = \Phi_{c_0} - \Phi_{c_m}.}$$

In particular, every closed path has zero total scalar response.

Proof. The identity telescopes:

$$\sum_{j=1}^{m} \left(\Phi_{c_{j-1}} - \Phi_{c_j}\right) = \Phi_{c_0} - \Phi_{c_m}.$$

□

Corollary 9.7 (Coxeter path relations)**.** *At multilinear degree four, the scalar connected response respects the adjacent-transposition relations*

$$s_i^2 = 1, \qquad s_i s_j = s_j s_i \quad (|i - j| > 1), \qquad s_i s_{i+1} s_i = s_{i+1} s_i s_{i+1}.$$

Thus involution loops, commuting squares, and braid paths carry no residual scalar obstruction.

Different paths may distribute their intermediate responses among different Jordan and nested-commutator channels, but their total scalar sum is forced to agree. The path dependence left open after section 8 therefore vanishes for the complete multilinear connected coefficient. This statement does not assert path independence for a truncated subset of channels or for unrelated history-valued observables.

9.5 Six-order reconstruction transform

Order the six trace coordinates and six potentials using the same vertex list. Let R be the reversal permutation matrix and J the all-ones matrix. Then definition 9.2 is

$$\mathbf{\Phi} = -\frac{1}{(1-q)^4} M(q) \mathbf{T},$$

where

$$M(q) = q^2 J + (q - q^2) I + (q^3 - q^2) R.$$

The space decomposes into three invariant sectors:

$$\mathbb{C}^6 = V_0 \oplus V_+ \oplus V_-,$$

where V_0 is the constant line, V_+ is the two-dimensional reversal-even zero-sum sector, and V_- is the three-dimensional reversal-odd sector.

Theorem 9.8 (Mode eigenvalues and generic reconstruction)**.** *The eigenvalues of $M(q)$ on these sectors are*

$$\lambda_0 = q(q^2+4q+1), \qquad \lambda_+ = q(1-q)^2, \qquad \lambda_- = q(1-q)(1+q),$$

with multiplicities $1, 2, 3$, respectively. Therefore the six order-class coefficients $\mathbf{\Phi}$ determine all six cyclic trace coordinates $\mathbf{T}$ whenever

$$q \notin \{0, 1, -1, -2+\sqrt{3}, -2-\sqrt{3}\}.$$

Proof. The matrices J and R commute. On V_0, J acts by 6 and R by 1. On V_+, J vanishes and R acts by 1. On V_-, J vanishes and R acts by -1. Substitution in $M(q)$ gives the three eigenvalues. The transform is invertible exactly when all three are nonzero and the normalization denominator is defined. □

At $q = -1$, the reversal-odd sector disappears, so orientation-sensitive trace information is lost. At the two roots of $q^2 + 4q + 1$, only the constant mode is lost. These exceptional spectral parameters belong to the finite transform itself and are not claims about physical operating points.

9.6 Boundary of the classification

The result closes the complete multilinear four-factor permutation problem and the scalar adjacent-swap path-cocycle problem. It does not classify repeated-variable coefficients, five-factor cyclic classes, dimension-specific trace identities, or infinite-dimensional determinant channels. Those remain separate gates.

10 Repeated-variable fourth-order filtration

The square-free four-factor sector distinguishes six cyclic order classes. Repeated variables produce a different filtration: the trace-word space shrinks, some classes become self-reversing, and connected coefficients can contain scalar lifts of lower-order channels. This section treats the two basic fourth-order multiplicity patterns

$$(2,1,1) \qquad \text{and} \qquad (2,2).$$

Throughout, coefficient extraction is formal. No global convergence theorem for a Magnus series is required.

10.1 The $(2,1,1)$ necklace sector

Let

$$U_{CBA} = e^{zC}e^{yB}e^{xA}, \qquad U_{CAB} = e^{zC}e^{xA}e^{yB},$$

and define the normalized connected generators

$$L_{CBA} = \log\frac{\det(I-qU_{CBA})}{(1-q)^d}, \qquad L_{CAB} = \log\frac{\det(I-qU_{CAB})}{(1-q)^d}.$$

At multidegree $(2,1,1)$, cyclic reduction leaves three trace necklaces:

$$T_+ = \operatorname{Tr}(A^2BC), \qquad T_- = \operatorname{Tr}(A^2CB), \qquad T_0 = \operatorname{Tr}(ABAC).$$

The first two form a reversal pair, while T_0 is self-reversing.

Theorem 10.1 (Repeated-letter order potentials)**.** *The connected coefficients are*

$$\boxed{[x^2yz]L_{CBA} = -\frac{1}{(1-q)^4}\left[\frac{q^2(1+q)}{2}T_+ + \frac{q(1+q)}{2}T_- + q^2T_0\right]}$$

and

$$\boxed{[x^2yz]L_{CAB} = -\frac{1}{(1-q)^4}\left[\frac{q(1+q)}{2}T_+ + \frac{q^2(1+q)}{2}T_- + q^2T_0\right].}$$

Consequently,

$$\boxed{[x^2yz](L_{CBA} - L_{CAB}) = \frac{q(1+q)}{2(1-q)^3}\operatorname{Tr}\big(A^2[B,C]\big).}$$

Proof. Set $r = q/(1-q)$ and write $U = I + W$. Only the terms

$$-\sum_{k=1}^{4}\frac{r^k}{k}\operatorname{Tr}(W^k)$$

can contribute to $[x^2yz]$. For the order CBA, cyclic word counting gives the coefficients

	r	r^2	r^3	r^4
T_-	$-\frac{1}{2}$	-2	$-\frac{5}{2}$	-1
T_+	0	$-\frac{1}{2}$	$-\frac{3}{2}$	-1
T_0	0	-1	-2	$-1.$

These sum respectively to

$$-\frac{q(1+q)}{2(1-q)^4}, \qquad -\frac{q^2(1+q)}{2(1-q)^4}, \qquad -\frac{q^2}{(1-q)^4}.$$

Swapping A and B exchanges the weights of T_+ and T_- while leaving the self-reversing class T_0 unchanged. Subtraction gives

$$T_+ - T_- = \operatorname{Tr}(A^2BC - A^2CB) = \operatorname{Tr}\big(A^2[B,C]\big).$$

□

Remark 10.2 (Spectral divided-power lift)**.** *If*

$$K_3(q) = \frac{q}{(1-q)^2}$$

is the cubic kernel from definition 6.1, then

$$\frac{q(1+q)}{2(1-q)^3} = \frac{1}{2}q\frac{d}{dq}K_3(q).$$

Thus the repeated-A channel is a divided-power spectral lift of the cubic kernel, but it carries the new trace insertion A^2.

10.2 Centered primitive channel

The insertion A^2 may itself contain a scalar lift of the cubic channel. Put

$$\overline{A} = \frac{\mathrm{Tr}(A)}{d}, \qquad A_0 = A - \overline{A}I.$$

Proposition 10.3 (Centered decomposition)**.** *One has*

$$\boxed{\mathrm{Tr}\big(A^2[B,C]\big) = \mathrm{Tr}\big(A_0^2[B,C]\big) + 2\overline{A}\ \mathrm{Tr}\big(A[B,C]\big).}$$

The first term is a centered repeated-letter channel, while the second is an inherited scalar lift of the cubic channel.

Proof. Expand $A = A_0 + \overline{A}I$. The term proportional to $\overline{A}^2$ vanishes because $\mathrm{Tr}([B,C]) = 0$, and

$$\mathrm{Tr}(A_0[B,C]) = \mathrm{Tr}(A[B,C]).$$

□

Example 10.4 (Repeated channel independent of the cubic channel)**.** *In* $\mathrm{M}_3(\mathbb{C})$*, take*

$$A = E_{12} + E_{23}, \qquad B = E_{32}, \qquad C = E_{21}.$$

Then $A^3 = 0$*,* $\mathrm{Tr}(A) = 0$*, and*

$$\mathrm{Tr}(A[B,C]) = 0, \qquad \mathrm{Tr}(A^2[B,C]) = 1.$$

Hence the cubic channel vanishes while the centered repeated-letter channel remains nonzero. Direct determinant expansion gives

$$[x^2yz](L_{CBA} - L_{CAB}) = \frac{q(1+q)}{2(1-q)^3}.$$

This proves that the centered $(2,1,1)$ *channel is not determined by the cubic channel alone.*

10.3 Connected versus disconnected order response

Let

$$F_\pi = \frac{\det(I - qU_\pi)}{(1-q)^d} = e^{L_\pi},$$

and write

$$\ell^\pi_{abc} = [x^a y^b z^c]L_\pi, \qquad f^\pi_{abc} = [x^a y^b z^c]F_\pi.$$

All proper lower coefficients are the same for the two orders CBA and CAB, except the cubic coefficient ℓ_{111}. Therefore the multivariate moment–cumulant relation reduces to

$$\boxed{\Delta f_{211} = \Delta\ell_{211} + \ell_{100}\,\Delta\ell_{111},}$$

where Δ means $CBA - CAB$. Since

$$\ell_{100} = -\frac{q}{1-q}\,\mathrm{Tr}(A),$$

we obtain

$$\boxed{\Delta f_{211} = \frac{q(1+q)}{2(1-q)^3}\,\mathrm{Tr}\big(A^2[B,C]\big) - \frac{q^2}{(1-q)^3}\,\mathrm{Tr}(A)\,\mathrm{Tr}\big(A[B,C]\big).}$$

The second term is a disconnected product of a first moment and the cubic order channel. Passing to the logarithm removes it from the primitive connected coefficient.

10.4 The balanced $(2,2)$ sector

Let

$$U_{BA} = e^{yB}e^{xA}, \qquad L_{BA} = \log \frac{\det(I - qU_{BA})}{(1-q)^d}.$$

Cyclic reduction of words with content A^2B^2 leaves two self-reversing necklaces:

$$S_1 = \mathrm{Tr}(A^2B^2), \qquad S_2 = \mathrm{Tr}(ABAB).$$

Theorem 10.5 (Balanced repeated-letter coefficient)**.** *For either order BA or AB,*

$$\boxed{[x^2y^2]L = -\frac{q(1+q)^2}{4(1-q)^4}\,\mathrm{Tr}(A^2B^2) - \frac{q^2}{2(1-q)^4}\,\mathrm{Tr}(ABAB).}$$

Equivalently,

$$\boxed{[x^2y^2]L = -\frac{q(q^2+4q+1)}{16(1-q)^4}\,\mathrm{Tr}\big(\{A,B\}^2\big) + \frac{q(q^2+1)}{16(1-q)^4}\,\mathrm{Tr}\big([A,B]^2\big).}$$

Proof. The cut-count coefficients of S_1 in powers r, r^2, r^3, r^4 are

$$-\frac{1}{4}, \quad -\frac{5}{4}, \quad -2, \quad -1,$$

while those of S_2 are

$$0, \quad -\frac{1}{2}, \quad -1, \quad -\frac{1}{2}.$$

Summing with $r = q/(1-q)$ gives the first formula. The second follows from

$$\mathrm{Tr}\big([A,B]^2\big) = 2(S_2 - S_1), \qquad \mathrm{Tr}\big(\{A,B\}^2\big) = 2(S_2 + S_1).$$

Exact equality of the BA and AB coefficients also follows from

$$\det(I - qe^{yB}e^{xA}) = \det(I - qe^{xA}e^{yB}).$$

□

The commutator-square trace is a connected noncommutative channel, but it is common to both factor orders. It measures internal noncommutativity without repairing two-factor spectral order blindness.

Example 10.6 (Order-neutral commutator square)**.** *In $\mathrm{M}_2(\mathbb{C})$, take $A = E_{12}$ and $B = E_{21}$. Then*

$$\mathrm{Tr}(A^2B^2) = 0, \qquad \mathrm{Tr}(ABAB) = 1, \qquad \mathrm{Tr}([A,B]^2) = 2.$$

The two characteristic determinants are identical and

$$[x^2y^2]L = -\frac{q^2}{2(1-q)^4}.$$

Thus a nonzero repeated connected coefficient need not be an order-discriminating coefficient.

10.5 Moment–cumulant filtration at $(2, 2)$

Write

$$L(x,y) = \sum_{\substack{a,b\ge 0\\ a+b>0}} \ell_{ab}x^a y^b, \qquad F(x,y) = e^{L(x,y)} = \sum_{a,b\ge 0} f_{ab}x^a y^b.$$

Then

$$\begin{aligned} f_{22} &= \ell_{22} + \ell_{20}\ell_{02} + \ell_{21}\ell_{01} + \ell_{12}\ell_{10} + \frac{1}{2}\ell_{11}^2 + \ell_{11}\ell_{10}\ell_{01} \\ &\quad + \frac{1}{2}\ell_{20}\ell_{01}^2 + \frac{1}{2}\ell_{02}\ell_{10}^2 + \frac{1}{4}\ell_{10}^2\ell_{01}^2. \end{aligned}$$

The single coefficient ℓ_{22} is the connected trace-word contribution. Every remaining term is a product of lower-degree connected coefficients and is disconnected in the determinant coefficient.

10.6 Multi-parameter reconstruction of repeated necklaces

Repeated-letter order experiments do not generally reconstruct all necklace coordinates at one spectral value. Spectral variation supplies the missing equations.

Theorem 10.7 (Two-parameter repeated-necklace reconstruction)**.** *Let $q_1, q_2 \neq 1$ satisfy*

$$q_1q_2(q_1 - q_2)(q_1q_2 - 1) \neq 0.$$

Then:

(i) the two values of $[x^2y^2]L(q)$ at q_1, q_2 determine both

$$\mathrm{Tr}(A^2B^2) \quad \textit{and} \quad \mathrm{Tr}(ABAB);$$

(ii) the two order potentials $[x^2yz]L_{CBA}$ and $[x^2yz]L_{CAB}$, sampled at q_1, q_2, determine all three necklaces

$$T_+, \quad T_-, \quad T_0.$$

Proof. For the $(2, 2)$ sector, after removing the common denominator and sign, the coefficient matrix has rows

$$\left(\frac{q_i(1+q_i)^2}{4}, \frac{q_i^2}{2}\right),$$

whose determinant is

$$\frac{1}{8}q_1q_2(q_1 - q_2)(q_1q_2 - 1).$$

For the $(2, 1, 1)$ sector, the order difference at any nonexceptional sampled value recovers $T_+ - T_-$. The order sum at q_i is

$$-\frac{1}{(1-q_i)^4}\left[\frac{q_i(1+q_i)^2}{2}(T_+ + T_-) + 2q_i^2T_0\right].$$

The determinant of these two even-sector rows is

$$q_1q_2(q_1 - q_2)(q_1q_2 - 1).$$

The stated condition therefore reconstructs the reversal-even sum and the self-reversing necklace, while the difference reconstructs the reversal-odd coordinate. □

The reciprocal degeneracy $q_1q_2 = 1$ reflects the identity

$$\frac{q}{(1+q)^2} = \frac{q^{-1}}{(1+q^{-1})^2}.$$

It is a property of the finite coefficient transform, not a physical symmetry assumption.

10.7 Boundary of the repeated-letter result

The section classifies the first two fourth-order repeated-variable sectors and separates connected coefficients from lower-degree determinant products. It does not yet classify the remaining multiplicity patterns $(3,1)$ and (4), dimension-specific trace identities, or repeated-variable channels at fifth and higher total degree.

11 Completion of the fourth-order multiplicity partition

The preceding sections classify the square-free sector, the repeated-letter sectors $(2,1,1)$ and $(2,2)$, and their connected versus disconnected filtration. Two multiplicity patterns remain:

$$(3,1) \qquad \text{and} \qquad (4).$$

They are simpler combinatorially because each has only one cyclic trace necklace in the declared model. Their coefficient kernel nevertheless carries the same fourth-order Eulerian polynomial that appeared in the constant mode of the six-class square-free transform.

Throughout this section, a multiplicity such as $(3,1)$ means that each distinct matrix label appears in one exponential and the repeated degree arises from Taylor powers of that exponential. A model with several separately positioned occurrences of the same matrix is a different split-occurrence problem and is not silently identified with the present one.

Define the fourth-order scalar kernel

$$\boxed{K_4(q) = -\frac{q(q^2+4q+1)}{24(1-q)^4}.}$$

11.1 The $(3,1)$ sector

Let

$$U_{BA} = e^{yB}e^{xA}, \qquad U_{AB} = e^{xA}e^{yB},$$

and define

$$L_{BA} = \log\frac{\det(I-qU_{BA})}{(1-q)^d}, \qquad L_{AB} = \log\frac{\det(I-qU_{AB})}{(1-q)^d}.$$

The content A^3B has one cyclic trace necklace,

$$R_{31} = \operatorname{Tr}(A^3B).$$

Theorem 11.1 (Mixed quartic coefficient)**.** *For either two-factor order,*

$$\boxed{[x^3y]L_{BA} = [x^3y]L_{AB} = -\frac{q(q^2+4q+1)}{6(1-q)^4}\operatorname{Tr}(A^3B).}$$

Equivalently,

$$[x^3y]L = 4K_4(q)\operatorname{Tr}(A^3B).$$

Proof. Set $r = q/(1-q)$ and write $U = I+W$. Only

$$-\sum_{k=1}^{4}\frac{r^k}{k}\operatorname{Tr}(W^k)$$

can contribute to $[x^3y]$. Since every word with content A^3B is cyclically equivalent, the four cut-count contributions multiplying $\operatorname{Tr}(A^3B)$ are

$$-\frac{r}{6}, \qquad -\frac{7r^2}{6}, \qquad -2r^3, \qquad -r^4.$$

Their sum is

$$-\frac{q(q^2+4q+1)}{6(1-q)^4}.$$

Equality of the two factor orders also follows directly from Sylvester's identity

$$\det(I-qe^{yB}e^{xA})=\det(I-qe^{xA}e^{yB}).$$

□

The factor four between $[x^3y]L$ and $[x^4]L$ below is the ordinary polarization multiplicity $4!/(3!1!)$. It is not a new spectral channel.

11.2 Centered filtration in the $(3,1)$ sector

Put

$$a=\frac{\mathrm{Tr}(A)}{d},\qquad b=\frac{\mathrm{Tr}(B)}{d},\qquad A_0=A-aI,\qquad B_0=B-bI.$$

Proposition 11.2 (Centered mixed quartic decomposition)**.** *One has*

$$\begin{aligned}\mathrm{Tr}(A^3B)=\;&\mathrm{Tr}(A_0^3B_0)+b\,\mathrm{Tr}(A_0^3)+3a\,\mathrm{Tr}(A_0^2B_0)+3ab\,\mathrm{Tr}(A_0^2)\\&+3a^2\,\mathrm{Tr}(A_0B_0)+da^3b.\end{aligned}$$

Thus $\mathrm{Tr}(A_0^3B_0)$ *is the fully centered mixed quartic coordinate, while the remaining terms are scalar lifts of lower centered moments.*

Proof. Expand

$$A^3B=(A_0+aI)^3(B_0+bI)$$

and use $\mathrm{Tr}(A_0)=\mathrm{Tr}(B_0)=0$. □

Example 11.3 (Primitive centered $(3,1)$ packet)**.** *In* $\mathrm{M}_4(\mathbb{C})$*, take*

$$A=\mathrm{diag}(-3,-1,1,3),\qquad B=\mathrm{diag}(-1,3,-3,1).$$

Then

$$\mathrm{Tr}(A)=\mathrm{Tr}(B)=0,\qquad \mathrm{Tr}(AB)=0,\qquad \mathrm{Tr}(A^2B)=0,$$

while

$$\mathrm{Tr}(A^3B)=48.$$

Hence the centered mixed quartic coordinate is not determined by the lower mixed moments. The matrices commute, so this example isolates a primitive repeated moment rather than an order-discriminating channel.

11.3 Moment–cumulant filtration at $(3,1)$

Write

$$L(x,y)=\sum_{\substack{a,b\ge 0\\a+b>0}}\ell_{ab}x^ay^b,\qquad F(x,y)=e^{L(x,y)}=\sum_{a,b\ge 0}f_{ab}x^ay^b.$$

Then

$$\boxed{\begin{aligned}f_{31}=\ell_{31}+\ell_{30}\ell_{01}+\ell_{21}\ell_{10}+\ell_{20}\ell_{11}+\frac{1}{2}\ell_{11}\ell_{10}^2\\+\ell_{20}\ell_{10}\ell_{01}+\frac{1}{6}\ell_{10}^3\ell_{01}.\end{aligned}}$$

Only ℓ_{31} is the connected fourth-order single-trace coefficient. Every other term is assembled from lower connected coefficients.

11.4 The pure (4) sector

For one generator set

$$U_A = e^{xA}, \qquad L_A = \log \frac{\det(I - qe^{xA})}{(1-q)^d}.$$

There is one cyclic trace coordinate,

$$R_4 = \mathrm{Tr}(A^4).$$

Theorem 11.4 (Pure quartic coefficient)**.** *The fourth connected coefficient is*

$$\boxed{[x^4]L_A = -\frac{q(q^2+4q+1)}{24(1-q)^4}\,\mathrm{Tr}(A^4) = K_4(q)\,\mathrm{Tr}(A^4).}$$

Proof. The four cut-count contributions multiplying $\mathrm{Tr}(A^4)$ are

$$-\frac{r}{24}, \qquad -\frac{7r^2}{24}, \qquad -\frac{r^3}{2}, \qquad -\frac{r^4}{4}.$$

Their sum at $r = q/(1-q)$ is $K_4(q)$. □

Let

$$a = \frac{\mathrm{Tr}(A)}{d}, \qquad A_0 = A - aI.$$

Then

$$\boxed{\mathrm{Tr}(A^4) = \mathrm{Tr}(A_0^4) + 4a\,\mathrm{Tr}(A_0^3) + 6a^2\,\mathrm{Tr}(A_0^2) + da^4.}$$

The first term is the centered quartic moment; the remaining terms are scalar lifts of lower centered moments.

Example 11.5 (Primitive centered quartic packet)**.** *In* $\mathrm{M}_4(\mathbb{C})$*, take*

$$A = \mathrm{diag}(1, -1, i, -i).$$

Then

$$\mathrm{Tr}(A) = \mathrm{Tr}(A^2) = \mathrm{Tr}(A^3) = 0, \qquad \mathrm{Tr}(A^4) = 4.$$

Thus the centered pure quartic moment is independent of all lower power sums in this exact complex packet.

The ordinary determinant coefficient obeys the one-variable cumulant identity

$$\boxed{f_{40} = \ell_{40} + \ell_{30}\ell_{10} + \frac{1}{2}\ell_{20}^2 + \frac{1}{2}\ell_{20}\ell_{10}^2 + \frac{1}{24}\ell_{10}^4.}$$

Again, only ℓ_{40} is the connected fourth-order single-trace term.

11.5 Fourth-order partition synthesis

The five integer partitions of total degree four determine five formal cyclic necklace sectors in the one-exponential-per-label model.

Theorem 11.6 (Fourth-order multiplicity synthesis)**.** *The formal fourth-order cyclic trace filtration has the following structure.*

Partition	***Necklaces***	***Order information***	***Generic reconstruction***
$(1,1,1,1)$	*6*	*full cyclic-order potential; reversal and adjacent-swap modes*	*six order coefficients at one admissible q*
$(2,1,1)$	*3*	*one reversal-odd order channel plus two even coordinates*	*both orders at two nonzero, distinct, nonreciprocal q-values*
$(2,2)$	*2*	*order-neutral in the collapsed two-factor model*	*two nonzero, distinct, nonreciprocal q-values*
$(3,1)$	*1*	*order-neutral*	*one q with* $K_4(q) \neq 0$
(4)	*1*	*order-neutral*	*one q with* $K_4(q) \neq 0$

Consequently, the formal necklace count is

$$6+3+2+1+1=13.$$

Within the declared collapsed-label model, the only fourth-order partition types that carry order-discriminating connected channels are

$$(1,1,1,1) \qquad \textit{and} \qquad (2,1,1).$$

The remaining three partitions contain connected noncommutative or centered moment information but no factor-order distinction.

For the one-necklace sectors $(3,1)$ and (4), reconstruction fails only at

$$q \in \{0, 1, -2+\sqrt{3}, -2-\sqrt{3}\},$$

where the kernel vanishes or the normalization is singular. The polynomial q^2+4q+1 is the same constant-mode factor that appears in the square-free six-order transform.

The synthesis theorem does not classify split-occurrence words in which several separately positioned exponentials carry the same matrix label. Nor does it impose dimension-specific trace identities that may reduce the formal necklace rank in small matrix dimensions.

12 General square-free cyclic-descent potential and fifth-order classification

The preceding four-factor calculations admit a general square-free formulation. For distinct matrix labels, the connected multilinear coefficient depends only on the relative cyclic descent number between the factor order and the trace word. The fifth-order theory is then a finite specialization with 24 oriented cyclic classes and a four-sector reconstruction transform.

12.1 Relative cyclic descents

Let

$$\mathcal{C}_n = S_n/C_n$$

denote the oriented cyclic classes of n distinct labels. For a source class

$$c = [c_1 c_2 \cdots c_n]_{\mathrm{cyc}}$$

choose the representative factor word

$$U_c(\mathbf{x}) = e^{x_{c_1} A_{c_1}} e^{x_{c_2} A_{c_2}} \cdots e^{x_{c_n} A_{c_n}}.$$

For a target cyclic class

$$e = [e_1 e_2 \cdots e_n]_{\mathrm{cyc}},$$

let $p_c(e_j)$ be the position of e_j in the source word c. Define

$$\mathrm{cdes}_c(e) = \#\left\{j : p_c(e_{j+1}) < p_c(e_j)\right\}, \qquad e_{n+1} = e_1.$$

This is independent of the cyclic representative of e. For distinct labels,

$$1 \le \mathrm{cdes}_c(e) \le n - 1.$$

The notions of cyclic descent and cyclic Eulerian algebra are classical [9]; the spectral weight below is the paper-specific bridge to connected log determinants.

For each target class define

$$T_e = \mathrm{Tr}(A_{e_1} A_{e_2} \cdots A_{e_n}),$$

and define the square-free connected potential

$$\Phi_c^{(n)}(q) = \left[\prod_{j=1}^{n} x_j\right] \log \frac{\det(I - qU_c(\mathbf{x}))}{(1-q)^d}.$$

Lemma 12.1 (Cyclic cut count)**.** *Fix source and target classes $c, e \in \mathcal{C}_n$, and write*

$$s = \mathrm{cdes}_c(e).$$

In the coefficient of T_e inside

$$-\frac{r^k}{k}\,\mathrm{Tr}(W^k), \qquad r = \frac{q}{1-q}, \qquad U_c = I + W,$$

the number of ordered k-block decompositions is

$$N_k(c, e) = \begin{cases} k\binom{n-s}{k-s}, & k \ge s, \\ 0, & k < s. \end{cases}$$

Proof. The multilinear part of W contains one word for every nonempty subsequence of the source order c. To produce the target cyclic word e from a product of k such subsequences, every relative cyclic descent of e must be a block boundary. There are s forced boundaries. The remaining $k - s$ boundaries are chosen among the $n - s$ ascent gaps, giving $\binom{n-s}{k-s}$ choices. Finally, the cyclic trace admits k choices of which block is written first. This gives the stated count. □

Theorem 12.2 (General square-free cyclic-descent potential)**.** *For every $c \in \mathcal{C}_n$,*

$$\boxed{\Phi_c^{(n)}(q) = -\frac{1}{(1-q)^n} \sum_{e \in \mathcal{C}_n} q^{\mathrm{cdes}_c(e)} T_e.}$$

Proof. For a target e with relative cyclic descent number s, definition 12.1 gives the total coefficient

$$-\sum_{k=s}^{n} \frac{r^k}{k} \left(k\binom{n-s}{k-s}\right) = -r^s (1+r)^{n-s}.$$

Since

$$r = \frac{q}{1-q}, \qquad 1 + r = \frac{1}{1-q},$$

this equals

$$-\frac{q^s}{(1-q)^n}.$$

Summing over all target cyclic classes proves the formula. □

Corollary 12.3 (General order difference and exact path potential)**.** *For any $c, d \in \mathcal{C}_n$,*

$$\Phi_c^{(n)} - \Phi_d^{(n)} = -\frac{1}{(1-q)^n} \sum_{e \in \mathcal{C}_n} \left(q^{\mathrm{cdes}_c(e)} - q^{\mathrm{cdes}_d(e)} \right) T_e.$$

If an oriented edge response is defined by

$$\omega(c, d) = \Phi_c^{(n)} - \Phi_d^{(n)},$$

then every path telescopes and every closed scalar path has zero total response.

The theorem places the three-, four-, and five-factor square-free calculations inside one cyclic-descent hierarchy. It does not claim that cyclic descents are new combinatorial objects; it identifies their exact rational spectral weight in the connected determinant channel.

12.2 The twenty-four fifth-order classes

For five distinct labels,

$$|\mathcal{C}_5| = \frac{5!}{5} = 24.$$

For every source class c, the relative cyclic descent distribution is

$$\#\{e : \mathrm{cdes}_c(e) = 1, 2, 3, 4\} = (1, 11, 11, 1).$$

The unique class with one relative cyclic descent is c itself, and the unique class with four relative cyclic descents is its reversal $\rho(c)$. Define

$$\mathcal{D}_j(c) = \{e \in \mathcal{C}_5 : \mathrm{cdes}_c(e) = j\}.$$

Theorem 12.4 (Fifth-order cyclic potential)**.** *For every $c \in \mathcal{C}_5$,*

$$\boxed{\Phi_c^{(5)}(q) = -\frac{1}{(1-q)^5} \left[qT_c + q^2 \sum_{e \in \mathcal{D}_2(c)} T_e + q^3 \sum_{e \in \mathcal{D}_3(c)} T_e + q^4 T_{\rho(c)} \right]}.$$

The coefficient layers are therefore

$$q, \qquad q^2, \qquad q^3, \qquad q^4,$$

with multiplicities

$$1, \qquad 11, \qquad 11, \qquad 1.$$

Example 12.5 (Exact five-cycle calibration)**.** *In $\mathrm{M}_5(\mathbb{C})$, set*

$$A = E_{12}, \quad B = E_{23}, \quad C = E_{34}, \quad D = E_{45}, \quad E = E_{51}.$$

For representatives with relative cyclic descent numbers $s = 1, 2, 3, 4$, direct determinant expansion gives

$$\det(I - qU_c) = (1-q)^5 - q^s abcde.$$

Thus the connected multilinear coefficient is exactly

$$-\frac{q^s}{(1-q)^5},$$

matching definition 12.4 in all four descent layers.

12.3 Local adjacent-swap spectator channels

Consider

$$U_{EDCBA} = e^{vE}e^{wD}e^{zC}e^{yB}e^{xA}, \qquad U_{EDCAB} = e^{vE}e^{wD}e^{zC}e^{xA}e^{yB}.$$

For a spectator word P in C, D, E, write

$$\mathrm{Tr}(A[B, P]) = \mathrm{Tr}(ABP - APB).$$

Theorem 12.6 (Fifth-order adjacent-swap context formula)**.** *The complete square-free adjacent-swap coefficient is*

$$\boxed{\begin{aligned}[xyzwv](L_{EDCBA} - L_{EDCAB}) = \frac{q^3}{(1-q)^4}\,\mathrm{Tr}(A[B, CDE]) + \frac{q}{(1-q)^4}\,\mathrm{Tr}(A[B, EDC]) \\ + \frac{q^2}{(1-q)^4} \sum_{P\in\{CED,DCE,DEC,ECD\}} \mathrm{Tr}(A[B, P]).\end{aligned}}$$

Proof. Apply definition 12.3 to the two source classes and group the resulting trace-word differences into the six formal context vectors $\mathrm{Tr}(A[B, P])$, $P \in S_3 \cdot CDE$. The direct spectator order CDE has weight q^3, its reversal EDC has weight q, and the four remaining spectator orders have weight q^2. One factor of $1 - q$ cancels from the difference of the two fifth-order potentials. □

The six context vectors are formally independent before dimension-specific trace identities are imposed. The theorem gives one weighted scalar response, not six independent measurements at one fixed order and one fixed value of q.

12.4 The fifth-order adjacent-swap graph

Let G_5 have vertex set $\mathcal{C}_5$, with an edge when two cyclic classes differ by one adjacent transposition, including the adjacent pair that crosses the chosen cyclic origin.

Theorem 12.7 (Fifth-order order graph)**.** *The graph G_5 has the following properties:*

(i) it has 24 vertices and is 5-regular;

(ii) it is connected and bipartite, with 12 vertices in each parity class;

(iii) its diameter is 4;

(iv) reversal is a fixed-point-free graph automorphism;

(v) the reversal quotient is the crown graph

$$K_{6,6} \setminus 6K_2;$$

(vi) its adjacency characteristic polynomial is

$$\boxed{(t^2 - 25)(t^2 - 1)^5(t^2 - 5)^6.}$$

Proof. There are $5!/5 = 24$ cyclic classes. Each class has five cyclically adjacent pairs, and swapping them gives five distinct neighbouring classes. Cyclic rotation by five positions has even parity, so parity is well defined on $\mathcal{C}_5$; every adjacent transposition reverses it. Connectivity follows because adjacent transpositions generate all linear permutations before taking the cyclic quotient.

The remaining finite statements follow from the exact canonical enumeration of the 24 classes. Reversal has no fixed class and preserves parity. Its quotient has two parts of size six and degree five, hence it is $K_{6,6}$ with one perfect matching removed. The same exact adjacency matrix gives the diameter and displayed characteristic polynomial. The complete enumeration is implemented in the accompanying verification script. □

Because the edge response is the difference of vertex potentials, all involution, commuting-square, and braid-path scalar residues vanish exactly at fifth-order square-free degree, just as in the four-factor case.

12.5 Fifth-order cyclic-descent transform

Order the 24 trace coordinates and potentials using the same cyclic-class list. For $j = 1, 2, 3, 4$, define the relation matrices

$$(D_j)_{c,e} = \begin{cases} 1, & \mathrm{cdes}_c(e) = j, \\ 0, & \text{otherwise.} \end{cases}$$

Then

$$D_1 = I, \qquad D_4 = R,$$

where R is reversal, and

$$\mathbf{\Phi} = -\frac{1}{(1-q)^5} M_5(q)\mathbf{T},$$

with

$$M_5(q) = qI + q^2 D_2 + q^3 D_3 + q^4 R.$$

Lemma 12.8 (Fifth cyclic-descent relation algebra)**.** *The relation matrices commute and satisfy*

$$D_2^2 = D_3^2 = 10I + 5D_2 + 5D_3 + R,$$

$$D_2 D_3 = I + 5D_2 + 5D_3 + 10R,$$

$$D_2 R = D_3, \qquad D_3 R = D_2, \qquad R^2 = I.$$

Proof. For fixed source and target classes, the matrix products count intermediate cyclic classes with prescribed pairs of relative cyclic descent numbers. Exact enumeration of the 24 classes gives the displayed intersection numbers. These are the fifth-order finite representation of the commutative cyclic Eulerian structure related to cyclic descent algebras [9]. □

Set

$$S = D_2 + D_3, \qquad H = D_2 - D_3.$$

Then

$$S^2 = 22(I + R) + 20S, \qquad H^2 = 18(I - R), \qquad SH = 0,$$

and

$$RS = S, \qquad RH = -H.$$

The 24-dimensional coordinate space therefore decomposes as

$$\mathbb{C}^{24} = V_0 \oplus V_+ \oplus V_-^{(+)} \oplus V_-^{(-)},$$

with dimensions

$$1, \qquad 11, \qquad 6, \qquad 6.$$

Here V_0 is the constant line, V_+ is the reversal-even zero-sum sector, and the two reversal-odd sectors are distinguished by

$$H = +6I \qquad \text{or} \qquad H = -6I.$$

Theorem 12.9 (Fifth-order mode eigenvalues and reconstruction)**.** *The eigenvalues of* $M_5(q)$ *on*

$$V_0, \quad V_+, \quad V_-^{(+)}, \quad V_-^{(-)}$$

are respectively

$$\boxed{\lambda_0 = q(1+q)(q^2+10q+1),}$$

$$\boxed{\lambda_+ = q(1-q)^2(1+q),}$$

$$\boxed{\lambda_-^{(+)} = q(1-q)(q^2+4q+1),}$$

$$\boxed{\lambda_-^{(-)} = q(1-q)^3.}$$

Their multiplicities are $1, 11, 6, 6$. *Consequently, the* 24 *fifth-order class potentials reconstruct all* 24 *cyclic trace coordinates whenever*

$$\boxed{q \notin \left\{0, 1, -1, -2 \pm \sqrt{3}, -5 \pm 2\sqrt{6}\right\}.}$$

Proof. On V_0, the row sums of D_2, D_3, R are $11, 11, 1$. On V_+, the corresponding eigenvalues are $-1, -1, 1$. On the two reversal-odd sectors they are

$$(3, -3, -1) \qquad \text{and} \qquad (-3, 3, -1).$$

Substitution into

$$M_5(q) = qI + q^2 D_2 + q^3 D_3 + q^4 R$$

gives the four displayed eigenvalues. The transform is invertible exactly when none vanishes and the normalization at $q = 1$ is defined. $\square$

At $q = -1$, both reversal-even sectors are lost while the two reversal-odd sectors remain. At $q = -2 \pm \sqrt{3}$, one six-dimensional reversal-odd sector is lost. At $q = -5 \pm 2\sqrt{6}$, only the constant mode is lost. These are exceptional values of the formal transform and are not physical operating claims.

12.6 Boundary of the fifth-order result

The section closes the square-free fifth-order cyclic-class potential, adjacent-swap graph, scalar path response, and generic 24-coordinate reconstruction problem. It does not classify repeated-variable fifth-order partitions, split-occurrence repeated events, dimension-specific trace identities, or infinite-dimensional determinant channels.

13 Fifth-order repeated-variable hierarchy and observability defects

The square-free fifth-order sector has 24 cyclic trace coordinates and is generically reconstructible from its 24 order potentials. Repeated multiplicities behave differently. Some repeated-word coordinates remain invisible even when every cyclic source order is sampled and the spectral parameter q is varied. This section classifies all remaining partitions of total degree five in the declared one-exponential-per-distinct-label model:

$$(2,1,1,1), \quad (2,2,1), \quad (3,1,1), \quad (3,2), \quad (4,1), \quad (5).$$

The theory of descent statistics on multipermutations is classical [10]. The present calculation concerns the different problem of the exact trace-word weights produced by the connected log determinant.

13.1 Two universal filtrations

The repeated-variable calculations use two elementary triangular decompositions.

Lemma 13.1 (Universal centering filtration)**.** *For every matrix label A_j, write*

$$a_j = \frac{\operatorname{Tr}(A_j)}{d}, \qquad A_{j,0} = A_j - a_j I.$$

For a trace word

$$T = \operatorname{Tr}(A_{i_1} A_{i_2} \cdots A_{i_5}),$$

one has

$$\boxed{T = \sum_{S \subseteq \{1,\dots,5\}} \left(\prod_{s \in S} a_{i_s} \right) \operatorname{Tr}\left(\overrightarrow{\prod_{t \notin S}} A_{i_t,0} \right).}$$

The arrow preserves the original cyclic order, and the trace of the empty product is d. The term with $S = \varnothing$ is the fully centered fifth-order coordinate; every other term is a scalar lift of a lower centered word.

Proof. Substitute $A_j = A_{j,0} + a_j I$, expand, and use that identity factors may be removed without changing the order of the remaining matrices. □

Lemma 13.2 (Universal multivariate moment–cumulant filtration)**.** *Let*

$$L(\mathbf{x}) = \sum_{\beta \in \mathbb{N}^m \setminus \{0\}} \ell_\beta \, \mathbf{x}^\beta, \qquad F(\mathbf{x}) = e^{L(\mathbf{x})} = \sum_{\alpha \in \mathbb{N}^m} f_\alpha \, \mathbf{x}^\alpha.$$

Then

$$\boxed{f_\alpha = \sum_{\substack{(m_\beta)_{\beta \neq 0} \\ \sum_\beta m_\beta \beta = \alpha}} \prod_{\beta \neq 0} \frac{\ell_\beta^{m_\beta}}{m_\beta!}.}$$

In particular, ℓ_α is the unique connected single-trace term at multidegree α; all other summands are disconnected products of lower connected coefficients.

Proof. Expand

$$e^L = \prod_{\beta \neq 0} \exp(\ell_\beta \mathbf{x}^\beta)$$

and collect the coefficient of $\mathbf{x}^\alpha$. □

13.2 The $(2,1,1,1)$ sector

Let the repeated label be A. The six cyclic source orders of the four distinct labels are

$$\mathcal{C}_4 = (ABCD, ABDC, ACBD, ACDB, ADBC, ADCB).$$

The twelve cyclic trace necklaces split into six adjacent-A words

$$\mathbf{U} = \begin{pmatrix} \operatorname{Tr}(A^2BCD) \\ \operatorname{Tr}(A^2BDC) \\ \operatorname{Tr}(A^2CBD) \\ \operatorname{Tr}(A^2CDB) \\ \operatorname{Tr}(A^2DBC) \\ \operatorname{Tr}(A^2DCB) \end{pmatrix}$$

and six separated-A words

$$\mathbf{V} = \begin{pmatrix} \mathrm{Tr}(ABACD) \\ \mathrm{Tr}(ABADC) \\ \mathrm{Tr}(ABCAD) \\ \mathrm{Tr}(ABDAC) \\ \mathrm{Tr}(ACADB) \\ \mathrm{Tr}(ACBAD) \end{pmatrix}.$$

Let R_U and R_V denote reversal on these two ordered lists, and set

$$M_4(q) = q^2 J + (q - q^2) I + (q^3 - q^2) R_U.$$

Define

$$X = \begin{pmatrix} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 \end{pmatrix}.$$

Theorem 13.3 ($(2,1,1,1)$ source-order transform)**.** *Let $\mathbf{\Phi}_{2111}(q) \in \mathbb{C}^6$ be the vector of connected $(2,1,1,1)$ coefficients for the six source orders in $\mathcal{C}_4$. Then*

$$\boxed{\mathbf{\Phi}_{2111}(q) = -\frac{1}{(1-q)^5}\left[\frac{1+q}{2} M_4(q)\mathbf{U} + q^2 X(I + qR_V)\mathbf{V}\right].}$$

Moreover,

$$XR_V = R_U X, \qquad \operatorname{rank} X = 4,$$

and the exact all-q blind space is

$$\boxed{\ker X = \operatorname{span}\left\{ \begin{pmatrix} -1 \\ -1 \\ 0 \\ 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ -1 \\ 1 \\ 0 \\ 0 \\ 1 \end{pmatrix} \right\}.}$$

Consequently, the complete source-order and spectral-parameter family has functional rank 10*, not* 12*.*

Proof. The adjacent-A words inherit the fourth-order own, reverse, and other weights, multiplied by the divided-power factor $(1+q)/2$. Direct cyclic cut counting of the separated words gives the matrix $q^2 X(I + qR_V)$. Exact row reduction gives the displayed rank and kernel. □

The observable separated-A coordinates may be chosen as

$$V_1 + \cdots + V_6, \qquad V_1 - V_2, \qquad V_3 - V_6, \qquad V_4 - V_5.$$

The two reversal-even contrasts

$$\boxed{\eta_1 = -V_1 - V_2 + V_4 + V_5, \qquad \eta_2 = -V_1 - V_2 + V_3 + V_6}$$

are invisible for every source order and every q.

Corollary 13.4 (Generic two-q reconstruction on the visible quotient)**.** *Two spectral values q_1, q_2 reconstruct all ten visible coordinates when*

$$q_i \notin \{0, 1, -1\}, \qquad q_1 \neq q_2, \qquad q_1 q_2 \neq 1.$$

The determinant factors controlling the constant and reversal-odd sectors are, up to nonzero normalization,

$$q_1 q_2 (q_1 + 1)(q_2 + 1)(q_1 - q_2)(q_1 q_2 - 1)$$

and

$$q_1 q_2 (q_1 - 1)(q_2 - 1)(q_1 - q_2)(q_1 q_2 - 1),$$

respectively.

13.3 The $(2,2,1)$ sector

Use the six necklaces

$$\begin{aligned} N_1 &= \mathrm{Tr}(A^2 B^2 C), \quad & N_2 &= \mathrm{Tr}(A^2 BCB), \quad & N_3 &= \mathrm{Tr}(A^2 C B^2), \\ N_4 &= \mathrm{Tr}(ABABC), \quad & N_5 &= \mathrm{Tr}(ABACB), \quad & N_6 &= \mathrm{Tr}(ABBAC). \end{aligned}$$

Reversal exchanges $N_1 \leftrightarrow N_3$ and $N_4 \leftrightarrow N_5$, while N_2, N_6 are self-reversing. Let Φ_+ and Φ_- denote the source orders ABC and ACB, and put

$$\Sigma = \frac{\Phi_+ + \Phi_-}{2}, \qquad \Delta = \frac{\Phi_+ - \Phi_-}{2}.$$

Theorem 13.5 ($(2,2,1)$ even and odd channels)**.** *One has*

$$\boxed{\Sigma = -\frac{1}{(1-q)^5} \left[\frac{q(1+q)^3}{8} (N_1 + N_3) + \frac{q^2(1+q)}{2} (N_2 + N_4 + N_5 + N_6) \right]}$$

and

$$\boxed{\Delta = -\frac{1}{(1-q)^4} \left[\frac{q(1+q)^2}{8} (N_1 - N_3) + \frac{q^2}{2} (N_4 - N_5) \right].}$$

Equivalently, the full order difference is

$$\boxed{\Phi_+ - \Phi_- = -\frac{1}{(1-q)^4} \left[\frac{q(1+q)^2}{4} \,\mathrm{Tr}\big(A^2 [B^2, C]\big) + q^2 \,\mathrm{Tr}\big(ABA[B, C]\big) \right].}$$

The all-q transform has rank 4. Its two-dimensional blind space may be represented by

$$\boxed{\beta_1 = N_6 - N_2, \qquad \beta_2 = N_4 + N_5 - 2N_2.}$$

The two order channels are independent. In $\mathrm{M}_3(\mathbb{C})$,

$$A = E_{11}, \qquad B = E_{12} + E_{22}, \qquad C = E_{21}$$

gives

$$\mathrm{Tr}(A^2[B^2, C]) = 1, \qquad \mathrm{Tr}(ABA[B, C]) = 0,$$

whereas

$$A = E_{12}, \qquad B = E_{21}, \qquad C = E_{11}$$

gives the values 0 and 1.

Corollary 13.6 (Two-q visible reconstruction)**.** *Two nonzero, distinct, nonreciprocal values q_1, q_2, with $q_i \notin \{-1, 1\}$, reconstruct the four visible coordinates. The even and odd reconstruction determinants are*

$$\frac{1}{16} q_1 q_2 (q_1 + 1)(q_2 + 1)(q_1 - q_2)(q_1 q_2 - 1)$$

and

$$\frac{1}{16} q_1 q_2 (q_1 - 1)(q_2 - 1)(q_1 - q_2)(q_1 q_2 - 1),$$

respectively.

13.4 The $(3, 1, 1)$ sector

Set

$$P_1 = \mathrm{Tr}(A^3 BC), \quad P_2 = \mathrm{Tr}(A^3 CB), \quad P_3 = \mathrm{Tr}(A^2 BAC), \quad P_4 = \mathrm{Tr}(A^2 CAB).$$

The pairs $P_1 \leftrightarrow P_2$ and $P_3 \leftrightarrow P_4$ are reversal pairs. Let

$$E_4(q) = q^2 + 4q + 1.$$

Theorem 13.7 ($(3, 1, 1)$ potential and exact blind pair)**.** *For source orders ABC and ACB,*

$$\boxed{\Phi_+ = -\frac{1}{(1-q)^5} \left[\frac{q E_4(q)}{6} P_1 + \frac{q^2 E_4(q)}{6} P_2 + \frac{q^2 (1+q)}{2} (P_3 + P_4) \right]}$$

and

$$\boxed{\Phi_- = -\frac{1}{(1-q)^5} \left[\frac{q^2 E_4(q)}{6} P_1 + \frac{q E_4(q)}{6} P_2 + \frac{q^2 (1+q)}{2} (P_3 + P_4) \right].}$$

Hence

$$\boxed{\Phi_+ - \Phi_- = -\frac{q E_4(q)}{6(1-q)^4} \mathrm{Tr}\big(A^3 [B, C]\big).}$$

The all-q transform has rank 3, and the reversal-odd coordinate

$$\boxed{P_3 - P_4}$$

is invisible for every q.

If

$$A = A_0 + aI, \qquad a = \frac{\mathrm{Tr}(A)}{d},$$

then

$$\boxed{\mathrm{Tr}(A^3[B, C]) = \mathrm{Tr}(A_0^3[B, C]) + 3a \,\mathrm{Tr}(A_0^2[B, C]) + 3a^2 \,\mathrm{Tr}(A[B, C]).}$$

The first term is a centered primitive fifth-order order channel. An exact packet in $\mathrm{M}_4(\mathbb{C})$ is

$$A = E_{12} + E_{23} + E_{34}, \qquad B = E_{42}, \qquad C = E_{21},$$

for which

$$\mathrm{Tr}(A[B, C]) = 0, \quad \mathrm{Tr}(A^2[B, C]) = 0, \quad \mathrm{Tr}(A^3[B, C]) = 1.$$

Two generic spectral values reconstruct $P_1, P_2, P_3 + P_4$, but no number of unmarked spectral samples reconstructs $P_3 - P_4$.

13.5 The order-neutral sectors

For content A^3B^2, define

$$Q_1 = \mathrm{Tr}(A^3B^2), \qquad Q_2 = \mathrm{Tr}(A^2BAB).$$

Both necklaces are self-reversing.

Theorem 13.8 ((3, 2) coefficient)**.** *For either two-factor order,*

$$\boxed{[x^3y^2]L = -\frac{1}{(1-q)^5}\left[\frac{q(1+q)(q^2+4q+1)}{12}Q_1 + \frac{q^2(1+q)}{2}Q_2\right].}$$

Two values q_1, q_2 reconstruct Q_1, Q_2 whenever

$$q_1q_2(q_1+1)(q_2+1)(q_1-q_2)(q_1q_2-1) \neq 0.$$

Define the pure fifth-order kernel

$$\boxed{K_5(q) = -\frac{q(1+q)(q^2+10q+1)}{120(1-q)^5}.}$$

Theorem 13.9 ((4, 1) and (5) coefficients)**.** *For either two-factor order,*

$$\boxed{[x^4y]L = 5K_5(q)\,\mathrm{Tr}(A^4B),}$$

while

$$\boxed{[x^5]L = K_5(q)\,\mathrm{Tr}(A^5).}$$

Both sectors are order-neutral.

With $A = A_0 + aI$, $B = B_0 + bI$,

$$\begin{aligned}\mathrm{Tr}(A^4B) = \;& \mathrm{Tr}(A_0^4B_0) + b\,\mathrm{Tr}(A_0^4) + 4a\,\mathrm{Tr}(A_0^3B_0) + 4ab\,\mathrm{Tr}(A_0^3) \\ & + 6a^2\,\mathrm{Tr}(A_0^2B_0) + 6a^2b\,\mathrm{Tr}(A_0^2) + 4a^3\,\mathrm{Tr}(A_0B_0) + da^4b,\end{aligned}$$

and

$$\mathrm{Tr}(A^5) = \mathrm{Tr}(A_0^5) + 5a\,\mathrm{Tr}(A_0^4) + 10a^2\,\mathrm{Tr}(A_0^3) + 10a^3\,\mathrm{Tr}(A_0^2) + da^5.$$

The exact packet

$$A = \mathrm{diag}(-2,-1,0,1,2), \qquad B = \mathrm{diag}(1,-4,6,-4,1)$$

satisfies

$$\mathrm{Tr}(A^kB) = 0 \quad (0 \le k \le 3), \qquad \mathrm{Tr}(A^4B) = 24.$$

For the pure sector, taking the five fifth roots of unity as the eigenvalues of A gives

$$\mathrm{Tr}(A^k) = 0 \quad (1 \le k \le 4), \qquad \mathrm{Tr}(A^5) = 5.$$

13.6 Fifth-order multiplicity synthesis

Theorem 13.10 (Fifth-order collapsed-label synthesis)**.** *Across all seven partitions of total degree five, including the square-free sector, the formal cyclic necklace counts, source-order counts, and all-q functional ranks are*

partition	*necklaces*	*source orders*	*functional rank*	*blind dimension*
$(1,1,1,1,1)$	24	24	24	0
$(2,1,1,1)$	12	6	10	2
$(2,2,1)$	6	2	4	2
$(3,1,1)$	4	2	3	1
$(3,2)$	2	1	2	0
$(4,1)$	1	1	1	0
(5)	1	1	1	0

and therefore

$$\boxed{24+12+6+4+2+1+1=50,}$$

while the combined functional rank is

$$\boxed{24+10+4+3+2+1+1=45.}$$

Thus exactly five formal fifth-order necklace contrasts remain spectrally blind in the collapsed-label model, even after all source orders and arbitrary spectral-parameter variation are allowed.

The order-discriminating fifth-order partitions are

$$(1,1,1,1,1), \qquad (2,1,1,1), \qquad (2,2,1), \qquad (3,1,1).$$

The two-factor and one-factor partitions

$$(3,2), \qquad (4,1), \qquad (5)$$

are order-neutral, although they retain connected centered moments and noncommutative trace information.

The five exact blind dimensions provide a direct bridge to the observer theory: unmarked spectral variation alone cannot close these fibres. Extra marked measurements must be introduced if a target depends on the missing repeated-word contrasts.

13.7 Boundary of the fifth-order multiplicity result

The classification is formal and finite-dimensional. It treats Taylor multiplicity inside one exponential per distinct label. It does not classify separately positioned repeated exponentials, impose dimension-specific trace identities, or prove an infinite-dimensional determinant extension.

14 General multiplicity transforms and split-occurrence lifting

The square-free cyclic-descent transform can be compressed systematically to any multiplicity partition. This gives a general derivation of the collapsed-label models treated at degrees four and five, and it identifies precisely which loss comes from target label identification and which loss comes from restricting the available source orders.

14.1 Occurrence polarization and target collapse

Let
$$\lambda = (m_1, \dots, m_r) \vdash n$$
be a multiplicity partition. Replace the m_i copies of the i-th matrix label by a distinct occurrence alphabet
$$\Omega_\lambda = \{(i,a) : 1 \le i \le r,\ 1 \le a \le m_i\}.$$
Let
$$\mathcal{C}(\Omega_\lambda)$$
denote the oriented cyclic classes of these n distinct occurrences, and let
$$\mathcal{N}_\lambda$$
denote the cyclic multiset necklaces of content λ. The collapse map
$$\pi_\lambda : \mathcal{C}(\Omega_\lambda) \longrightarrow \mathcal{N}_\lambda$$
forgets the occurrence index a while retaining the matrix label i.

Define the incidence matrix
$$(P_\lambda)_{e,N} = \begin{cases} 1, & \pi_\lambda(e) = N, \\ 0, & \text{otherwise.} \end{cases}$$
Its columns have disjoint supports, so
$$\operatorname{rank} P_\lambda = |\mathcal{N}_\lambda|.$$

For occurrence classes $c, e \in \mathcal{C}(\Omega_\lambda)$, let
$$(M_n(q))_{c,e} = q^{\mathrm{cdes}_c(e)}.$$
This is the square-free cyclic-descent matrix from definition 12.2.

Theorem 14.1 (Split-occurrence multiplicity transform)**.** *After identifying all occurrence matrices (i,a) with the same matrix A_i, the vector of split-occurrence connected coefficients is*
$$\boxed{\mathbf{\Psi}_\lambda(q) = -\frac{1}{(1-q)^n} M_n(q) P_\lambda\, \mathbf{T}_\lambda,}$$
where $\mathbf{T}_\lambda$ is the vector of multiset-necklace trace coordinates.

Equivalently, for a split source class c,
$$\boxed{\Psi_{c,\lambda}(q) = -\frac{1}{(1-q)^n} \sum_{N \in \mathcal{N}_\lambda} \left(\sum_{\substack{e \in \mathcal{C}(\Omega_\lambda) \\ \pi_\lambda(e) = N}} q^{\mathrm{cdes}_c(e)} \right) T_N.}$$

Proof. Apply the general square-free theorem on the occurrence alphabet Ω_λ. Once the occurrence matrices are identified, all square-free trace coordinates in the fibre $\pi_\lambda^{-1}(N)$ become the same multiset trace T_N. Summing the corresponding columns gives the incidence product $M_n(q)P_\lambda$. $\square$

14.2 Collapsed one-exponential-per-label transform

For a cyclic order σ of the r distinct matrix labels, let $b_\sigma \in \mathcal{C}(\Omega_\lambda)$ be the occurrence word obtained by placing the m_i occurrences of each label i consecutively inside its block. Let B_λ be the row selector whose rows are the block source classes b_σ, and put

$$d_\lambda = \prod_{i=1}^{r} m_i!.$$

Theorem 14.2 (Polarized collapsed-label transform)**.** *The connected coefficient of the monomial*

$$x_1^{m_1} \cdots x_r^{m_r}$$

in the one-exponential-per-distinct-label model is

$$\boxed{\mathbf{\Phi}_\lambda(q) = -\frac{1}{d_\lambda (1-q)^n} B_\lambda M_n(q) P_\lambda \, \mathbf{T}_\lambda.}$$

Proof. Replace

$$e^{x_i A_i}$$

by the adjacent split product

$$e^{x_{i,1}A_i} \cdots e^{x_{i,m_i}A_i} = e^{(x_{i,1}+\cdots+x_{i,m_i})A_i}.$$

For any formal series depending on $x_{i,1} + \cdots + x_{i,m_i}$, the coefficient of $\prod_a x_{i,a}$ is $m_i!$ times the coefficient of $x_i^{m_i}$. Dividing the split coefficient by $\prod_i m_i!$ and restricting to block source classes gives the formula. ☐

Write

$$M_n(q) = \sum_{s=1}^{n-1} q^s D_s, \qquad (D_s)_{c,e} = \mathbf{1}_{\{\mathrm{cdes}_c(e)=s\}}.$$

Corollary 14.3 (All-spectral-value rank and blind kernel)**.** *The functional rank of the collapsed multiplicity transform, allowing every block source order and arbitrary variation of q, is*

$$\boxed{R_\lambda = \operatorname{rank} \begin{pmatrix} B_\lambda D_1 P_\lambda \\ B_\lambda D_2 P_\lambda \\ \vdots \\ B_\lambda D_{n-1} P_\lambda \end{pmatrix}.}$$

Its exact spectral blind space is

$$\boxed{K_\lambda = \bigcap_{s=1}^{n-1} \ker(B_\lambda D_s P_\lambda),}$$

with

$$\dim K_\lambda = |\mathcal{N}_\lambda| - R_\lambda.$$

The factorial normalization and the common denominator do not affect rank. This gives a finite exact algorithm for every multiplicity partition $\lambda \vdash n$.

14.3 Generic completeness of split occurrences

Theorem 14.4 (Generic split-occurrence completeness). *For every n,*

$$M_n(q) = qI + O(q^2).$$

Consequently, $\det M_n(q)$ *is not the zero polynomial. For every multiplicity partition $\lambda \vdash n$, the split-occurrence map*

$$M_n(q)P_\lambda$$

has full column rank

$$|\mathcal{N}_\lambda|$$

for all q outside a finite exceptional set. Thus all multiset-necklace trace coordinates are reconstructible from all split source orders at one generic spectral value.

Proof. For a source class c, the only target class with one relative cyclic descent is c itself. Every off-diagonal target has at least two relative cyclic descents. Hence the coefficient of q in $M_n(q)$ is the identity. Therefore $\det M_n(q) = q^{|\mathcal{C}_n|} + O(q^{|\mathcal{C}_n|+1})$ and is not identically zero.

Away from its finite zero set, $M_n(q)$ is invertible. Since P_λ has full column rank,

$$\operatorname{rank}(M_n(q)P_\lambda) = \operatorname{rank} P_\lambda = |\mathcal{N}_\lambda|.$$

□

The collapsed blind kernel is therefore not caused by identifying equal target matrices alone. It is caused by restricting the source module to block words and dividing out the occurrence variables before observing their relative positions.

14.4 Minimal lifting of the five degree-five blind modes

At total degree five, the collapsed hierarchy has

$$50$$

formal necklaces and functional rank

$$45.$$

Let

$$K_5 = K_{2111} \oplus K_{221} \oplus K_{311}$$

be its five-dimensional blind space.

Use the blind basis

$$(\eta_1, \eta_2, \beta_2, \beta_1, \kappa),$$

where

$$\eta_1 = -V_1 - V_2 + V_4 + V_5, \qquad \eta_2 = -V_1 - V_2 + V_3 + V_6,$$
$$\beta_1 = N_6 - N_2, \qquad \beta_2 = N_4 + N_5 - 2N_2,$$

and

$$\kappa = P_4 - P_3.$$

Consider the following five split source classes, with repeated occurrences declared separately:

$$\begin{aligned} s_1 &= [A_1\, B\, A_2\, C\, D]_{\mathrm{cyc}}, \\ s_2 &= [A_1\, B\, C\, A_2\, D]_{\mathrm{cyc}}, \\ t_1 &= [A_1\, A_2\, B_1\, C\, B_2]_{\mathrm{cyc}}, \\ t_2 &= [A_1\, B_1\, A_2\, B_2\, C]_{\mathrm{cyc}}, \\ u_1 &= [A_1\, A_2\, B\, A_3\, C]_{\mathrm{cyc}}. \end{aligned}$$

Theorem 14.5 (Minimal five-probe split lift)**.** *At $q = 2$, after removing the common normalization factor, the restrictions of these five split coefficients to the ordered blind basis above form*

$$\boxed{G_{\mathrm{split}} = \begin{pmatrix} -6 & -6 & 0 & 0 & 0 \\ 0 & 6 & 0 & 0 & 0 \\ 0 & 0 & -12 & -6 & 0 \\ 0 & 0 & 6 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 \end{pmatrix}}.$$

Therefore

$$\det G_{\mathrm{split}} = -2592 \neq 0.$$

The five split probes lift all five collapsed blind modes.

Moreover, five is minimal: no family of fewer than five scalar supplemental measurements can make the collapsed degree-five map injective.

Proof. Exact cyclic-descent enumeration gives the displayed response matrix. Its nonzero determinant proves that the five restrictions are independent on K_5.

For minimality, let

$$G : \mathbb{C}^{50} \longrightarrow \mathbb{C}^m$$

be any supplemental linear observation map. Injectivity of the combined map requires $G|_{K_5}$ to be injective. Since $\dim K_5 = 5$, rank-nullity forces $m \geq 5$. □

Corollary 14.6 (Minimal observer criterion on the blind sector)**.** *A bank of five scalar linear observer functionals repairs the collapsed degree-five spectral map exactly when its restriction to K_5 has rank five. The split probes in definition 14.5 provide one explicit minimal bank.*

This corollary is a theorem about the declared finite trace-coordinate model. It does not by itself specify a physical marker, hardware placement, or protected execution architecture.

14.5 Boundary

The multiplicity transform above is general in n and in the partition λ. The split-completeness theorem proves that the degree-five 45/50 defect is repairable by enriching source occurrence information. It does not imply that endpoint measurements recover a factor history after different histories have already collapsed to the same endpoint. The trace-coordinate blind kernel and the endpoint-history kernel remain distinct information losses.

15 Complete marker banks and exact endpoint reconstruction

The Hilbert–Schmidt pairing on $\mathrm{M}_d(\mathbb{C})$ is nondegenerate. We use analysis vectors M_a with responses

$$\langle M_a, D \rangle_{\mathrm{HS}} = \mathrm{Tr}(M_a^* D).$$

If the same response is written as a physical marked trace $\mathrm{Tr}(N_a D)$, then the insertion marker is $N_a = M_a^*$. This adjoint convention is harmless but must be kept explicit when matrix units are indexed.

15.1 Matrix-unit bank

Let E_{ij} denote the standard matrix units.

Theorem 15.1 (Matrix-unit completeness)**.** *The Hilbert–Schmidt analysis bank*

$$\mathcal{M}_{\mathrm{unit}} = \{E_{ij} : 1 \le i, j \le d\}$$

is complete on $\mathrm{M}_d(\mathbb{C})$*. Indeed,*

$$\langle E_{ij}, D\rangle_{\mathrm{HS}} = \mathrm{Tr}(E_{ji}D) = D_{ij}.$$

Therefore

$$\mathcal{A}_{\mathcal{M}_{\mathrm{unit}}}(D) = 0 \quad \Longleftrightarrow \quad D = 0.$$

Proof. The product $E_{ji}D$ has trace D_{ij}. All entries are therefore recovered. □

Corollary 15.2. *For every nonzero* $D \in \mathrm{M}_d(\mathbb{C})$*, there exists an analysis vector* M *satisfying*

$$\langle M, D\rangle_{\mathrm{HS}} \neq 0.$$

Equivalently, the insertion marker $N = M^*$ *satisfies* $\mathrm{Tr}(ND) \neq 0$*.*

15.2 Hermitian bank

Physical or numerical measurement rules may prefer Hermitian markers. For $i < j$, set

$$S_{ij} = E_{ij} + E_{ji}, \qquad A_{ij} = i(E_{ij} - E_{ji}).$$

Theorem 15.3 (Hermitian completeness)**.** *The* d^2 *Hermitian matrices*

$$E_{ii}, \qquad S_{ij}, \qquad A_{ij}, \quad i < j,$$

form a complete bank on $\mathrm{M}_d(\mathbb{C})$*.*

Proof. Because the markers are Hermitian, their Hilbert–Schmidt responses equal their marked trace responses. They are

$$\mathrm{Tr}(E_{ii}D) = D_{ii},$$
$$\mathrm{Tr}(S_{ij}D) = D_{ji} + D_{ij},$$

and

$$\mathrm{Tr}(A_{ij}D) = i(D_{ji} - D_{ij}).$$

Hence

$$D_{ij} = \frac{\mathrm{Tr}(S_{ij}D) + i\,\mathrm{Tr}(A_{ij}D)}{2},$$
$$D_{ji} = \frac{\mathrm{Tr}(S_{ij}D) - i\,\mathrm{Tr}(A_{ij}D)}{2}.$$

All entries are reconstructed. □

15.3 Model-relative completeness

A bank with fewer than d^2 scalar measurements cannot be injective on the full d^2-dimensional complex matrix space. It may nevertheless be complete on a declared lower-dimensional residue model $\mathcal{D}$.

Proposition 15.4 (Subspace completeness)**.** *For a linear subspace* $\mathcal{D} \subseteq \mathrm{M}_d(\mathbb{C})$*, the bank* $\mathcal{M}$ *is complete on* $\mathcal{D}$ *exactly when*

$$\mathcal{D} \cap \mathrm{span}(\mathcal{M})^{\perp} = \{0\},$$

where orthogonality is Hilbert–Schmidt.

This formulation is useful when symmetries, sparsity, block structure, or a known endpoint model reduce the admissible residue space. The reduction must be declared independently; a small bank is not complete merely because its observed examples happen to be separated.

16 Frame bounds and noisy reconstruction

Exact completeness is binary. Robust observability is quantitative.

Let

$$\mathcal{A}_{\mathcal{M}}(D) = (\langle M_a, D\rangle_{\mathrm{HS}})_{a=1}^{r},$$

and define the frame operator

$$S_{\mathcal{M}} = \mathcal{A}_{\mathcal{M}}^{*}\mathcal{A}_{\mathcal{M}}, \qquad S_{\mathcal{M}}D = \sum_{a=1}^{r}\langle M_a, D\rangle_{\mathrm{HS}}M_a.$$

Let

$$\alpha = \lambda_{\min}(S_{\mathcal{M}}), \qquad \beta = \lambda_{\max}(S_{\mathcal{M}}).$$

Theorem 16.1 (Marker frame bounds)**.** *For every $D \in \mathrm{M}_d(\mathbb{C})$,*

$$\boxed{\alpha\|D\|_{\mathrm{HS}}^2 \le \|\mathcal{A}_{\mathcal{M}}D\|_2^2 \le \beta\|D\|_{\mathrm{HS}}^2.}$$

The bank is complete on the full matrix space if and only if $\alpha > 0$.

Proof. Since

$$\|\mathcal{A}_{\mathcal{M}}D\|_2^2 = \langle D, S_{\mathcal{M}}D\rangle_{\mathrm{HS}},$$

the inequalities follow from the extremal eigenvalues of the positive operator $S_{\mathcal{M}}$. Injectivity is equivalent to $\ker S_{\mathcal{M}} = \{0\}$. $\square$

The singular values of $\mathcal{A}_{\mathcal{M}}$ lie between $\sqrt{\alpha}$ and $\sqrt{\beta}$. Thus

$$\kappa(\mathcal{A}_{\mathcal{M}}) = \sqrt{\frac{\beta}{\alpha}}$$

when $\alpha > 0$.

16.1 Canonical reconstruction

Assume $\alpha > 0$. Define the canonical dual reconstruction

$$\mathcal{R}_{\mathcal{M}} = S_{\mathcal{M}}^{-1}\mathcal{A}_{\mathcal{M}}^{*}.$$

Then

$$\mathcal{R}_{\mathcal{M}}\mathcal{A}_{\mathcal{M}}D = D.$$

Suppose the observed data are

$$y = \mathcal{A}_{\mathcal{M}}D + \eta, \qquad \|\eta\|_2 \le \varepsilon,$$

and set

$$\widehat{D} = \mathcal{R}_{\mathcal{M}}y.$$

Theorem 16.2 (Sharp noisy reconstruction bound)**.**

$$\boxed{\|\widehat{D} - D\|_{\mathrm{HS}} \le \frac{\varepsilon}{\sqrt{\alpha}}.}$$

Proof. The pseudoinverse norm is the reciprocal of the smallest singular value:

$$\|\mathcal{R}_{\mathcal{M}}\| = \frac{1}{\sqrt{\alpha}}.$$

Therefore

$$\|\widehat{D} - D\|_{\mathrm{HS}} = \|\mathcal{R}_{\mathcal{M}}\eta\|_{\mathrm{HS}} \le \frac{\|\eta\|_2}{\sqrt{\alpha}}.$$

$\square$

16.2 Sharpness

Consider the scaled matrix-unit bank on $\mathrm{M}_2(\mathbb{C})$:

$$\{\delta E_{11}, E_{12}, E_{21}, E_{22}\}, \qquad \delta = \frac{1}{10}.$$

Then

$$\alpha = \frac{1}{100}, \qquad \beta = 1, \qquad \kappa = 10.$$

Noise of size $1/100$ in the weakest response produces reconstruction error $1/10$. Hence the bound in definition 16.2 is attained.

17 Robust discrimination and marker-bank design

17.1 Pairwise discrimination

Let two residues D_1, D_2 be measured independently:

$$y_k = \mathcal{A}_{\mathcal{M}} D_k + \eta_k, \qquad \|\eta_k\|_2 \le \varepsilon.$$

Then

$$y_1 - y_2 = \mathcal{A}_{\mathcal{M}}(D_1 - D_2) + (\eta_1 - \eta_2),$$

with

$$\|\eta_1 - \eta_2\|_2 \le 2\varepsilon.$$

Theorem 17.1 (Sharp 2ε threshold)**.** *If*

$$\boxed{\|\mathcal{A}_{\mathcal{M}}(D_1 - D_2)\|_2 > 2\varepsilon,}$$

then the two noisy observation sets cannot coincide. A sufficient native-space condition is

$$\boxed{\sqrt{\alpha}\|D_1 - D_2\|_{\mathrm{HS}} > 2\varepsilon.}$$

Proof. The reverse triangle inequality gives

$$\|y_1 - y_2\|_2 \ge \|\mathcal{A}_{\mathcal{M}}(D_1 - D_2)\|_2 - 2\varepsilon.$$

The lower frame inequality yields the second condition. □

At equality, opposite admissible noise vectors may erase the observed difference. Without additional probabilistic or structural assumptions on the noise, the strict threshold is optimal.

17.2 Marker perturbations

Suppose the implemented markers are

$$\widetilde{M}_a = M_a + \Delta M_a$$

and define

$$\mu = \left(\sum_a \|\Delta M_a\|_{\mathrm{HS}}^2\right)^{1/2}.$$

Then

$$\|\mathcal{A}_{\widetilde{\mathcal{M}}} - \mathcal{A}_{\mathcal{M}}\| \le \mu.$$

Theorem 17.2 (Perturbation stability)**.** *If $\mathcal{M}$ has frame bounds α, β, then*

$$(\sqrt{\alpha}-\mu)^2\|D\|_{\mathrm{HS}}^2 \le \|\mathcal{A}_{\widetilde{\mathcal{M}}}D\|_2^2 \le (\sqrt{\beta}+\mu)^2\|D\|_{\mathrm{HS}}^2$$

whenever $\mu < \sqrt{\alpha}$. In particular, completeness survives under the same strict condition.

Proof. For unit D,

$$\|\mathcal{A}_{\widetilde{\mathcal{M}}}D\|_2 \ge \|\mathcal{A}_{\mathcal{M}}D\|_2 - \|(\mathcal{A}_{\widetilde{\mathcal{M}}} - \mathcal{A}_{\mathcal{M}})D\|_2 \ge \sqrt{\alpha}-\mu.$$

The upper bound is analogous. □

17.3 Tight-bank optimality

Let

$$N = \dim_{\mathbb{C}} \mathrm{M}_d(\mathbb{C}) = d^2$$

and fix the total marker energy

$$E_{\mathcal{M}} = \sum_a \|M_a\|_{\mathrm{HS}}^2.$$

Since

$$\mathrm{Tr}(S_{\mathcal{M}}) = E_{\mathcal{M}},$$

the average frame eigenvalue is $E_{\mathcal{M}}/N$.

Theorem 17.3 (Energy optimality)**.** *Every bank satisfies*

$$\boxed{\alpha \le \frac{E_{\mathcal{M}}}{N} \le \beta.}$$

At fixed total marker energy, no bank can have lower frame bound larger than $E_{\mathcal{M}}/N$. Equality holds exactly when

$$S_{\mathcal{M}} = \frac{E_{\mathcal{M}}}{N}I,$$

that is, when the bank is tight.

Full-space completeness forces $r \ge N$. When $r \ge N$, tight banks attain the displayed optimum; when $r < N$, the lower frame bound is necessarily zero. Under the normalization $E_{\mathcal{M}} = N$, a tight bank has

$$\alpha = \beta = 1, \qquad \kappa = 1.$$

Thus tightness equalizes response strength across all matrix directions. Completeness alone prevents exact blindness; tightness prevents a weakest direction from becoming unnecessarily fragile.

18 Target faithfulness and blind-kernel classification

Full endpoint reconstruction may be unnecessary. A representation can lose matrix information while retaining everything required for a declared target.

Let X be a state set, let

$$T : X \to Y$$

be a representation, and let

$$\Pi : X \to Z$$

be a target map. Define the representation-fibre relation

$$x \approx_T x' \iff T(x) = T(x'),$$

and the target-fibre relation

$$x \approx_\Pi x' \iff \Pi(x) = \Pi(x').$$

Theorem 18.1 (Target-faithfulness factorization)**.** *The following are equivalent:*

(i) $T(x) = T(x')$ *implies* $\Pi(x) = \Pi(x')$*;*

(ii) $\approx_T \subseteq \approx_\Pi$*;*

(iii) there exists a unique map

$$\widehat{\Pi} : T(X) \to Z$$

such that

$$\Pi = \widehat{\Pi} \circ T.$$

Proof. If representation fibres lie inside target fibres, define

$$\widehat{\Pi}(Tx) = \Pi(x).$$

The fibre condition makes the definition independent of the representative. The converse follows from any factorization. □

18.1 Linear form

Let $A : V \to W$ and $P : V \to Z$ be linear.

Corollary 18.2 (Linear blind-kernel criterion)**.** *The target P factors through A if and only if*

$$\boxed{\ker A \subseteq \ker P.}$$

For a marker bank on a residue model V, this becomes

$$\bigcap_a \ker\langle M_a, \cdot\rangle \subseteq \ker P.$$

The bank can therefore be target-faithful without being complete on V.

18.2 Blindness classes

The kernel criterion separates two forms of information loss.

Target-irrelevant blindness. The representation identifies distinct states, but every identified pair has the same target value. The target remains computable.

Target-destroying blindness. There exist x, x' with

$$T(x) = T(x'), \qquad \Pi(x) \neq \Pi(x').$$

No post-processing of T can recover Π on all states.

Applied to characteristic determinants, the theorem says that ordinary spectral data may be sufficient for a spectral target while being unfaithful for an endpoint-order target. Applied to a marker bank, it says that a bank may be intentionally designed only for a protected target class rather than for full matrix tomography.

18.3 Stable target recovery

In finite-dimensional linear settings, exact kernel inclusion is only the algebraic gate. Stable target recovery also requires a lower bound on the representation over the quotient directions relevant to the target. The frame lower bound α is precisely this margin for a complete observer bank.

19 Endpoint reconstruction versus history reconstruction

A complete marker bank reconstructs the endpoint matrix. It does not reconstruct a factor history that has already collapsed to that endpoint.

Theorem 19.1 (Endpoint-history no-go). *Let $\mathfrak{W}$ be a class of factor words and let*

$$\mathcal{E} : \mathfrak{W} \to \mathrm{M}_d(\mathbb{C})$$

map each word to its endpoint product. Let $\mathcal{O}$ be any observation that factors through the endpoint:

$$\mathcal{O} = \widetilde{\mathcal{O}} \circ \mathcal{E}.$$

If two words w_1, w_2 satisfy

$$\mathcal{E}(w_1) = \mathcal{E}(w_2),$$

then

$$\mathcal{O}(w_1) = \mathcal{O}(w_2).$$

Hence no endpoint-only observation can separate all histories unless the endpoint map itself is injective on the declared word class.

Proof. Immediate from the factorization:

$$\mathcal{O}(w_1) = \widetilde{\mathcal{O}}(\mathcal{E}(w_1)) = \widetilde{\mathcal{O}}(\mathcal{E}(w_2)) = \mathcal{O}(w_2).$$

□

Example 19.2 (Same endpoint, different words). *Let $N^2 = 0$. Then*

$$(I + N)(I - N) = I = I \cdot I.$$

The two factor histories differ, but every endpoint marker produces the same response because the endpoint matrices are identical.

This motivates two distinct blindness types.

Definition 19.3 (Type A: endpoint spectral blindness). *Two endpoints differ but share the selected spectral shadow:*

$$U \neq V, \qquad \mathfrak{S}(U) = \mathfrak{S}(V).$$

A suitable marker bank can repair this blindness.

Definition 19.4 (Type B: erased-history blindness). *Two factor histories differ but have the same endpoint:*

$$w_1 \neq w_2, \qquad \mathcal{E}(w_1) = \mathcal{E}(w_2).$$

No endpoint-only marker bank can repair this blindness.

Type B requires additional state that is not a function of the final endpoint, for example authenticated prefix states, internal-position observations, a path lift, or a declared transcript. The present paper does not construct a universal history representation. It proves why one is necessary.

19.1 Minimal augmentation principle

Suppose a target depends on history through a map

$$\Pi_{\mathrm{hist}} : \mathfrak{W} \to Z.$$

Any sufficient representation $T : \mathfrak{W} \to Y$ must satisfy

$$\ker_{\mathrm{fib}} T \subseteq \ker_{\mathrm{fib}} \Pi_{\mathrm{hist}},$$

in the fibre sense of definition 18.1. Endpoint data alone are sufficient exactly when Π_{hist} is constant on endpoint fibres.

20 Exact finite models

The abstract results are illustrated by exact nilpotent matrices. These examples are small enough to verify symbolically and sharp enough to expose the relevant information loss.

20.1 Two-factor spectral collision repaired by a marker

On $\mathbb{C}^3$, let

$$X = I + E_{12}, \qquad Y = I + E_{23}.$$

Because $E_{12}E_{23} = E_{13}$ while $E_{23}E_{12} = 0$,

$$XY = I + E_{12} + E_{23} + E_{13},$$

$$YX = I + E_{12} + E_{23},$$

and therefore

$$YX - XY = -E_{13} \neq 0.$$

Nevertheless XY and YX are similar, so their characteristic determinants coincide.

Take

$$M = E_{31}.$$

Then

$$\mathrm{Tr}(M(YX - XY)) = -1.$$

Thus the marker detects a difference erased by the ordinary characteristic determinant. Moreover,

$$\det(I - qMYX) - \det(I - qMXY) = q.$$

The distinction appears already in the linear coefficient of the marked determinant.

A Hermitian alternative is

$$M_H = E_{13} + E_{31},$$

which also has nonzero trace pairing with the residue.

20.2 Three-factor noncyclic spectral channel

Let

$$A = E_{12}, \qquad B = E_{23}, \qquad C = E_{31}.$$

Then $A^2 = B^2 = C^2 = 0$ and

$$\chi(A, B, C) = \mathrm{Tr}(A[B, C]) = 1.$$

The exponentials truncate:

$$e^{xA} = I + xA, \quad e^{yB} = I + yB, \quad e^{zC} = I + zC.$$

Direct calculation gives

$$\det(I - qU_{CBA}) = (1-q)^3 - q^2xyz,$$

whereas

$$\det(I - qU_{CAB}) = (1-q)^3 - qxyz.$$

Consequently,

$$[xyz](L_{CBA} - L_{CAB}) = \frac{q}{(1-q)^2},$$

in agreement with definition 6.1.

20.3 Incomplete-bank blindness

A diagonal marker bank satisfies

$$\mathrm{Tr}(MD) = 0$$

for every purely off-diagonal residue D. Thus even a large collection of diagonal markers is blind on the off-diagonal subspace. The word "observer" does not imply completeness.

20.4 Sharp conditioning model

The scaled matrix-unit bank

$$\left\{ \frac{1}{10}E_{11}, E_{12}, E_{21}, E_{22} \right\}$$

is complete but has lower frame bound $1/100$. It reconstructs exactly in the absence of noise yet amplifies the weakest response by a factor of ten. This separates algebraic completeness from robust observability.

21 Interpretation, applications, and scope boundaries

The theorems are stated for finite matrices, but the underlying distinction appears whenever a process has an ordered internal evolution and a compressed endpoint summary.

21.1 What the theory says

> If a declared representation is many-to-one on the distinctions required by a target, no downstream decision rule using only that representation can be universally correct. Additional observations must reduce the blind fibres, and their robustness is controlled by a quantitative lower separation margin.

For distinct labels, square-free connected coefficients are classified by a relative cyclic-descent potential. For a general multiplicity partition, the collapsed repeated-variable coefficient is a factorial-normalized row restriction of the square-free occurrence transform. At total degree five the collapsed model has fifty formal necklaces and rank forty-five, but the full split-occurrence source module is generically complete. Five explicit split source coefficients lift the five-dimensional collapsed blind kernel, and five is the minimum possible number of scalar supplemental observations. This repair concerns missing endpoint trace coordinates; it does not repair a factor history already collapsed to the same endpoint.

21.2 What the theory does not say

The paper does not claim:

(i) that every noncommuting process is spectrally distinguishable;

(ii) that one marker is sufficient for every residue class;

(iii) that spectral variation inside the collapsed model closes its repeated-word blind modes;

(iv) that a complete endpoint bank reconstructs factor history;

(v) that the five abstract supplemental functionals automatically specify physical markers or hardware locations;

(vi) that a nonzero commutator-square trace is automatically an order marker;

(vii) that finite matrix markers automatically define gauge-invariant observables;

(viii) that a finite result extends to trace-class or Fredholm determinants without new hypotheses;

(ix) that a mathematical observer bank by itself constitutes a hardware architecture or a security system.

These exclusions keep endpoint and order-channel theorems separate from implementation claims and prevent a finite calibration from being promoted into an infinite-dimensional or physical conclusion.

21.3 Relation to the earlier thermodynamic paper

The earlier thermodynamic response manuscript [1] treated cyclic return residue, memory, and curvature in a state-variable setting. The current paper supplies a separate algebraic result: spectral summaries of ordered matrix products can erase order, while connected trace channels and declared observer measurements recover different layers of the lost endpoint information. The earlier work motivates the question but is not used as a proof dependency for the finite theorems.

21.4 Potential mathematical application classes

The finite theory may be relevant to:

- ordered transfer matrices and holonomy approximations;
- matrix product models and noncommutative signal summaries;
- endpoint verification in finite state-transition systems;
- cyclic-descent and multiset-necklace classification of ordered exponential coefficients;
- tomography of declared trace-word or endpoint residue classes;
- experimental design across source occurrence patterns and spectral parameters;
- feature-map sufficiency and information-loss analysis.

Any domain-specific claim requires an independent model identifying the endpoint matrix, admissible residue class, marker implementation, noise norm, spectral parameter regime, and protected target.

22 Open problems

The square-free connected potential is available at arbitrary distinct-label degree, the collapsed multiplicity transform is available for every partition, and generic split-occurrence completeness is proved. Several natural extensions remain.

Open Problem 22.1 (Dimension-specific trace identities). *The formal cyclic-word space may collapse under polynomial trace identities in small matrix dimensions. Determine quotient ranks and exceptional reconstruction parameters for each declared dimension, beginning with* $\mathrm{M}_2(\mathbb{C})$*,* $\mathrm{M}_3(\mathbb{C})$*, and* $\mathrm{M}_4(\mathbb{C})$*.*

Open Problem 22.2 (Symmetry-reduced minimal marker design). *The abstract degree-five defect space has dimension five, and five scalar functionals are necessary and sufficient. Classify marker banks that respect declared reversal, permutation, adjoint, or covariance symmetries while remaining injective on this defect space and maximizing the lower frame bound.*

Open Problem 22.3 (Robust split-occurrence design). *Among all minimal five split source probes, optimize conditioning under source constraints, perturbation, and noise. Determine whether a tight or nearly tight five-probe frame exists on the degree-five blind kernel.*

Open Problem 22.4 (Sparse and low-rank residues). *When the admissible endpoint difference belongs to a sparse, low-rank, or structured nonlinear model, determine the smallest marker count and sharp stable-recovery constants.*

Open Problem 22.5 (Probabilistic noise). *Replace adversarial bounds by stochastic noise models and derive concentration, testing, and minimax design results. The deterministic* 2ε *threshold should appear as a worst-case boundary.*

Open Problem 22.6 (Infinite-dimensional marker frames). *Let* $\mathcal{H}$ *be a Hilbert or trace-ideal space of operators. Determine conditions under which a countable family of trace-compatible markers forms a frame with a positive lower bound and supports stable reconstruction.*

Finite injectivity does not survive completion automatically. Closed range, topology, domain, and trace duality must be specified.

Open Problem 22.7 (Fredholm marked determinants). *For a trace-class or regularized operator family* $K(z)$*, determine hypotheses under which*

$$\det(I - qMK(z))$$

is well defined, holomorphic in the relevant variables, and faithful for a declared target. The choice of ordinary versus regularized determinant must be explicit.

Open Problem 22.8 (History-faithful representations). *For a declared class of factor words or paths, construct minimal additional observations that separate the history fibres relevant to a target. Endpoint markers alone cannot solve the problem when endpoint collisions exist.*

Open Problem 22.9 (Continuous-time order memory). *Connect the finite product results to time-ordered exponentials under explicit Magnus convergence or approximation hypotheses. One must separate formal coefficients from global convergence statements.*

23 Conclusion

Ordered evolution, endpoint matrices, spectral summaries, split source occurrences, and marked responses are different information layers. This paper proves exact boundaries between them and gives a finite recovery theory.

For distinct labels, the square-free connected potential is

$$\Phi_c^{(n)}(q) = -\frac{1}{(1-q)^n} \sum_{e \in \mathcal{C}_n} q^{\mathrm{cdes}_c(e)} T_e.$$

For a multiplicity partition $\lambda = (m_1, \ldots, m_r) \vdash n$, occurrence polarization gives

$$\mathbf{\Psi}_\lambda(q) = -\frac{1}{(1-q)^n} M_n(q) P_\lambda \mathbf{T}_\lambda,$$

and the collapsed one-exponential transform is

$$\mathbf{\Phi}_\lambda(q) = -\frac{1}{(\prod_i m_i!)(1-q)^n} B_\lambda M_n(q) P_\lambda \mathbf{T}_\lambda.$$

This formula supplies the exact all-parameter rank and blind kernel for every multiplicity partition.

At fifth order the collapsed ranks sum to forty-five across fifty formal necklaces. The five missing contrasts are not caused by identifying equal target matrices alone: $M_n(q) = qI + O(q^2)$, so the full split-occurrence map is generically complete. Five explicit split probes at $q = 2$ restrict to a blind-sector matrix with determinant -2592, and dimension counting proves that no fewer than five scalar supplemental observations can restore injectivity.

This source-occurrence repair is not the same as history recovery. The five modes are missing endpoint trace coordinates that can be restored by richer source or marked observations. History blindness occurs earlier: different factor histories may already have the same endpoint, in which case no endpoint-only marker can distinguish them.

Observer markers enlarge the representation. A nonzero trace pairing detects a spectrally invisible endpoint residue; complete finite banks reconstruct endpoint differences; and frame bounds convert completeness into quantitative robustness. Blind-kernel inclusion separates endpoint reconstruction from target preservation.

The resulting hierarchy is

$$\boxed{\text{ordered history} \longrightarrow \text{endpoint} \longrightarrow \text{spectral shadow} \longrightarrow \text{split or declared marked response.}}$$

Information may be lost at each compression. Cyclic-descent and multiplicity transforms characterize the spectral loss, split occurrences or marked banks can close declared endpoint defects, and repairing erased history requires additional history-bearing state.

A Proof of the cubic connected coefficient

We prove definition 6.1. Let

$$U = I + W.$$

Then

$$I - qU = (1-q)\left(I - \frac{q}{1-q} W\right).$$

Set

$$r = \frac{q}{1-q}.$$

The normalized connected determinant is

$$L = \operatorname{Tr}\log(I - rW) = -\sum_{k \geq 1} \frac{r^k}{k} \operatorname{Tr}(W^k).$$

Only $k = 1, 2, 3$ can contribute to the multilinear coefficient xyz.

Expand the two products

$$U_{CBA} = e^{zC}e^{yB}e^{xA}, \qquad U_{CAB} = e^{zC}e^{xA}e^{yB}$$

through total degree three. When the xyz coefficients in

$$\mathrm{Tr}(W), \qquad \mathrm{Tr}(W^2), \qquad \mathrm{Tr}(W^3)$$

are collected and trace words are reduced cyclically, the difference becomes

$$r(1+r)\big(\mathrm{Tr}(ABC) - \mathrm{Tr}(ACB)\big).$$

Since

$$r(1+r) = \frac{q}{(1-q)^2}$$

and

$$\mathrm{Tr}(ABC) - \mathrm{Tr}(ACB) = \mathrm{Tr}(A[B,C]),$$

we obtain

$$[xyz]\big(L_{CBA} - L_{CAB}\big) = \frac{q}{(1-q)^2}\,\mathrm{Tr}(A[B,C]).$$

The proof is formal in x, y, z, q and requires no global convergence theorem.

B Exact calibration ledger

This appendix records the principal finite exact models in a form suitable for independent symbolic verification.

Gate	Model	Exact result
Two-factor blindness	$X = I + E_{12}$, $Y = I + E_{23}$	$YX - XY = -E_{13} \neq 0$, while $\det(I - qYX) = \det(I - qXY)$.
Marked separation	$M = E_{31}$	$\mathrm{Tr}(M(YX - XY)) = -1$ and the marked determinant difference equals q.
Three-factor channel	$A = E_{12}, B = E_{23}, C = E_{31}$	$\mathrm{Tr}(A[B,C]) = 1$, with cubic connected difference $q/(1-q)^2$.
Complete bank	$\{E_{ji}\}_{i,j=1}^{d}$	$\mathrm{Tr}(E_{ji}D) = D_{ij}$.
Ill-conditioning	$\{0.1E_{11}, E_{12}, E_{21}, E_{22}\}$	$\alpha = 0.01$, $\beta = 1$, $\kappa = 10$; the noise bound is attained.
History collision	$N^2 = 0$	$(I+N)(I-N) = I = I \cdot I$.

These examples are theorem calibrations, not empirical evidence for any physical implementation.

C Reproducibility and source lineage

The mathematical source lineage for this manuscript is maintained in the private MP repository through the following theorem packages:

(1) connected determinant cumulants;

(2) Magnus–log-determinant order memory;

(3) observer-marked spectral separation;

(4) noisy observer-marker conditioning;

(5) cut representation faithfulness and reconstruction.

The companion project files `MP_SOURCE_CROSSWALK.md` and `PROJECT_MANIFEST.json` record the corresponding branches and pull requests.

References

[1] M. Dabas, Geometric Completion of Thermodynamic Response: From the T–V–S–P Compass to Recursive λ-Jacobian Geometry, *arXiv preprint* arXiv:2603.20773, 2026. doi:10.48550/arXiv.2603.20773.

[2] W. Magnus, On the exponential solution of differential equations for a linear operator, *Communications on Pure and Applied Mathematics* 7(4) (1954), 649–673. doi:10.1002/cpa.3160070404.

[3] S. Blanes, F. Casas, J. A. Oteo, and J. Ros, The Magnus expansion and some of its applications, *Physics Reports* 470(5–6) (2009), 151–238. doi:10.1016/j.physrep.2008.11.001.

[4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, second edition, Cambridge University Press, 2013.

[5] N. J. Higham, *Functions of Matrices: Theory and Computation*, Society for Industrial and Applied Mathematics, 2008.

[6] C. Procesi, The invariant theory of $n \times n$ matrices, *Advances in Mathematics* 19(3) (1976), 306–381. doi:10.1016/0001-8708(76)90027-X.

[7] U. Leron, Trace identities and polynomial identities of $n \times n$ matrices, *Journal of Algebra* 42(2) (1976), 369–377. doi:10.1016/0021-8693(76)90104-6.

[8] C. Reutenauer, *Free Lie Algebras*, London Mathematical Society Monographs, New Series 7, Oxford University Press, 1993. doi:10.1093/oso/9780198536796.001.0001.

[9] T. K. Petersen, Cyclic descents and P-partitions, *Journal of Algebraic Combinatorics* 22(3) (2005), 343–375. doi:10.1007/s10801-005-4532-5.

[10] S.-M. Ma, J. Ma, and Y.-N. Yeh, Eulerian polynomials for multipermutations and signed multipermutations, *arXiv preprint* arXiv:1907.13082, 2019.

[11] R. J. Duffin and A. C. Schaeffer, A class of nonharmonic Fourier series, *Transactions of the American Mathematical Society* 72 (1952), 341–366. doi:10.1090/S0002-9947-1952-0047179-6.

[12] O. Christensen, *An Introduction to Frames and Riesz Bases*, second edition, Birkhäuser, 2016.

[13] G. M. D'Ariano, P. Perinotti, and M. F. Sacchi, Informationally complete measurements and group representation, *Journal of Optics B: Quantum and Semiclassical Optics* 6 (2004), S487–S491. doi:10.1088/1464-4266/6/6/005.

[14] A. J. Scott, Tight informationally complete quantum measurements, *Journal of Physics A: Mathematical and General* 39(43) (2006), 13507–13530. doi:10.1088/0305-4470/39/43/009.